

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414852
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Watanabe; Yoshitaka et al.

Intraocular lens injector

Abstract

There is provided an intraocular lens injector for injecting an intraocular lens into an eye, including: an injector main body **5**; an injection tube **7** attached to a tip part of the injector main body **5** and having a nozzle portion **7b**; and an attachment member **16** having a protruding surface disposed in a state of protruding to outside from an outer circumferential surface of the nozzle portion **7b**, wherein the attachment member **16** has an engaging portion **16b** engaged with a sleeve portion **16a** and is configured to be movable with respect to the injection tube **7**.

Inventors:	Watanabe; Yoshitaka (Tokyo, JP), Zhang; Xiaoxiao (Singapore, SG)
Applicant:	Hoya Medical Singapore Pte. Ltd. (Singapore, SG)
Family ID:	1000008820489
Assignee:	Hoya Medical Singapore Pte. Ltd. (Singapore, SG)
Appl. No.:	16/313180
Filed (or PCT Filed):	June 28, 2017
PCT No.:	PCT/JP2017/023748
PCT Pub. No.:	WO2018/003853
PCT Pub. Date:	January 04, 2018

Prior Publication Data

Document Identifier	Publication Date
US 20190192284 A1	Jun. 27, 2019

Foreign Application Priority Data

JP	2016-127294	Jun. 28, 2016
----	-------------	---------------

Publication Classification

Int. Cl.: **A61F2/16** (20060101); **A61F9/007** (20060101); **A61M5/315** (20060101); **A61M5/32** (20060101); **A61M5/31** (20060101)

U.S. Cl.:

CPC **A61F2/1678** (20130101); **A61F2/16** (20130101); **A61F2/1662** (20130101); **A61F2/1672** (20130101); **A61F9/007** (20130101); **A61M5/31528** (20130101); **A61M5/3286** (20130101); **A61F2002/1683** (20130101); **A61M2005/3125** (20130101)

Field of Classification Search

CPC: A61F (2/1678); A61F (2/1672); A61F (2/1662); A61F (2/16); A61F (2/167); A61F (9/00736); A61M (5/31528); A61M (5/3286); A61M (2005/3125)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2761446	12/1955	Reed	N/A	N/A
3212685	12/1964	Swan	N/A	N/A
4205747	12/1979	Gilliam et al.	N/A	N/A
4269307	12/1980	LaHaye	N/A	N/A
4423809	12/1983	Mazzocco	N/A	N/A
4573998	12/1985	Mazzocco	N/A	N/A
4608049	12/1985	Kelman	N/A	N/A
4634423	12/1986	Bailey	N/A	N/A
4681102	12/1986	Bartell	N/A	N/A
4697697	12/1986	Graham et al.	N/A	N/A
4699140	12/1986	Holmes	N/A	N/A
4702244	12/1986	Mazzocco	N/A	N/A
4715373	12/1986	Mazzocco et al.	N/A	N/A
4747404	12/1987	Jampel et al.	N/A	N/A
4750498	12/1987	Graham	N/A	N/A
4759359	12/1987	Willis et al.	N/A	N/A
4763650	12/1987	Hauser	N/A	N/A
4765329	12/1987	Cumming et al.	N/A	N/A
4769034	12/1987	Poley	N/A	N/A
4781719	12/1987	Kelman	N/A	N/A
4787904	12/1987	Severin	N/A	N/A
4810249	12/1988	Haber et al.	N/A	N/A
4819631	12/1988	Poley	N/A	N/A
4834094	12/1988	Patton	N/A	N/A
4836201	12/1988	Patton	N/A	N/A
4862885	12/1988	Cumming	N/A	N/A
4880000	12/1988	Holmes et al.	N/A	N/A
4919130	12/1989	Stoy	606/107	A61F 2/1664

4934363	12/1989	Smith et al.	N/A	N/A
4955889	12/1989	Van Gent	N/A	N/A
4976716	12/1989	Cumming	N/A	N/A
4988352	12/1990	Poley	N/A	N/A
4994028	12/1990	Leonard et al.	N/A	N/A
5066297	12/1990	Cumming	N/A	N/A
5098439	12/1991	Hill et al.	N/A	N/A
5123905	12/1991	Kelman	N/A	N/A
5139501	12/1991	Klaas	N/A	N/A
5171241	12/1991	Buboltz et al.	N/A	N/A
5176686	12/1992	Poley	N/A	N/A
5178622	12/1992	Lehner	N/A	N/A
5190552	12/1992	Kelman	N/A	N/A
5190553	12/1992	Kanert et al.	N/A	N/A
5222972	12/1992	Hill et al.	N/A	N/A
5242450	12/1992	McDonald	N/A	N/A
5259395	12/1992	Li	N/A	N/A
5275604	12/1993	Rheinish et al.	N/A	N/A
5281227	12/1993	Sussman	N/A	N/A
5304182	12/1993	Rheinish et al.	N/A	N/A
5354333	12/1993	Kammann et al.	N/A	N/A
5395378	12/1994	McDonald	N/A	N/A
5425734	12/1994	Blake	N/A	N/A
5454818	12/1994	Hambleton et al.	N/A	N/A
5468246	12/1994	Blake	N/A	N/A
5474562	12/1994	Orchowski et al.	N/A	N/A
5494484	12/1995	Feingold	N/A	N/A
5496328	12/1995	Nakajima et al.	N/A	N/A
5499987	12/1995	Feingold	N/A	N/A
5562676	12/1995	Brady et al.	N/A	N/A
5571113	12/1995	McDonald	N/A	N/A
5578042	12/1995	Cumming	N/A	N/A
5582613	12/1995	Brady	N/A	N/A
5582614	12/1995	Feingold	N/A	N/A
5584304	12/1995	Brady	N/A	N/A
5616148	12/1996	Eagles et al.	N/A	N/A
5620450	12/1996	Eagles et al.	N/A	N/A
5643275	12/1996	Blake	N/A	N/A
5643276	12/1996	Zaleski	N/A	N/A
5645534	12/1996	Chanoch	N/A	N/A
5653715	12/1996	Reich et al.	N/A	N/A
5653753	12/1996	Brady et al.	N/A	N/A
5702402	12/1996	Brady	N/A	N/A
5702441	12/1996	Zhou	N/A	N/A
5716364	12/1997	Makker et al.	N/A	N/A
5728075	12/1997	Levander	N/A	N/A
5728102	12/1997	Feingold et al.	N/A	N/A
5735858	12/1997	Makker et al.	N/A	N/A
5766181	12/1997	Chambers et al.	N/A	N/A
5772666	12/1997	Feingold et al.	N/A	N/A

5772667	12/1997	Blake	N/A	N/A
5776138	12/1997	Vidal et al.	N/A	N/A
5800442	12/1997	Wolf et al.	N/A	N/A
5803925	12/1997	Yang et al.	N/A	N/A
5807400	12/1997	Chambers et al.	N/A	N/A
5810833	12/1997	Brady et al.	N/A	N/A
5810834	12/1997	Heyman	N/A	N/A
5860984	12/1998	Chambers et al.	N/A	N/A
5860986	12/1998	Reich et al.	N/A	N/A
5868751	12/1998	Feingold	N/A	N/A
5868752	12/1998	Makker et al.	N/A	N/A
5873879	12/1998	Figueroa et al.	N/A	N/A
5876406	12/1998	Wolf et al.	N/A	N/A
5876407	12/1998	Makker et al.	N/A	N/A
5876440	12/1998	Feingold	N/A	N/A
5891152	12/1998	Feingold	N/A	N/A
5902307	12/1998	Feingold et al.	N/A	N/A
5919197	12/1998	McDonald	N/A	N/A
5921989	12/1998	Deacon et al.	N/A	N/A
5928245	12/1998	Wolf et al.	N/A	N/A
5941886	12/1998	Feingold	N/A	N/A
5942277	12/1998	Makker et al.	N/A	N/A
5944725	12/1998	Cicenas	N/A	N/A
5947974	12/1998	Brady et al.	N/A	N/A
5947975	12/1998	Kikuchi et al.	N/A	N/A
5957748	12/1998	Ichiha	N/A	N/A
5957896	12/1998	Bendek et al.	N/A	N/A
6001107	12/1998	Feingold	N/A	N/A
6010510	12/1999	Brown et al.	N/A	N/A
6022358	12/1999	Wolf et al.	N/A	N/A
6048348	12/1999	Chambers et al.	N/A	N/A
6050999	12/1999	Paraschac et al.	N/A	N/A
6051000	12/1999	Heyman	N/A	N/A
6056757	12/1999	Feingold et al.	N/A	N/A
6056758	12/1999	Vidal et al.	N/A	N/A
6059791	12/1999	Chambers	N/A	N/A
6074397	12/1999	Chambers et al.	N/A	N/A
6083230	12/1999	Makker et al.	N/A	N/A
6093193	12/1999	Makker et al.	N/A	N/A
6129733	12/1999	Brady et al.	N/A	N/A
6142999	12/1999	Brady et al.	N/A	N/A
6143000	12/1999	Feingold	N/A	N/A
6162229	12/1999	Feingold et al.	N/A	N/A
6174315	12/2000	Chambers et al.	N/A	N/A
6214015	12/2000	Reich et al.	N/A	N/A
6241737	12/2000	Feingold	N/A	N/A
6248111	12/2000	Glick et al.	N/A	N/A
6251114	12/2000	Farmer et al.	N/A	N/A
6254607	12/2000	Makker et al.	N/A	N/A
6267768	12/2000	Deacon	N/A	N/A

6283975	12/2000	Glick et al.	N/A	N/A
6283976	12/2000	Portney	N/A	N/A
6312433	12/2000	Butts	N/A	N/A
6334862	12/2001	Vidal et al.	N/A	N/A
6336932	12/2001	Figueroa et al.	N/A	N/A
6355046	12/2001	Kikuchi et al.	N/A	N/A
6371960	12/2001	Heyman et al.	N/A	N/A
6386357	12/2001	Egawa	N/A	N/A
6387101	12/2001	Butts et al.	N/A	N/A
6398788	12/2001	Makker et al.	N/A	N/A
6406481	12/2001	Feingold et al.	N/A	N/A
6428545	12/2001	Portney	N/A	N/A
6447519	12/2001	Brady et al.	N/A	N/A
6447520	12/2001	Ott et al.	N/A	N/A
6468282	12/2001	Kikuchi et al.	N/A	N/A
6471708	12/2001	Green	N/A	N/A
6491697	12/2001	Clark et al.	N/A	N/A
6497708	12/2001	Cumming	N/A	N/A
6500181	12/2001	Portney	N/A	N/A
6503275	12/2002	Cumming	N/A	N/A
6506195	12/2002	Chambers et al.	N/A	N/A
6537283	12/2002	Van Noy	N/A	N/A
6540754	12/2002	Brady	N/A	N/A
6554839	12/2002	Brady	N/A	N/A
6558395	12/2002	Hjertman et al.	N/A	N/A
6605093	12/2002	Blake	606/107	A61F 2/1664
6607537	12/2002	Binder	N/A	N/A
6629979	12/2002	Feingold	N/A	N/A
6666871	12/2002	Kikuchi et al.	N/A	N/A
6679891	12/2003	Makker et al.	N/A	N/A
6685740	12/2003	Figueroa et al.	N/A	N/A
6712848	12/2003	Wolf et al.	N/A	N/A
6723104	12/2003	Ott	N/A	N/A
6733507	12/2003	McNicholas et al.	N/A	N/A
6749631	12/2003	Pietrini et al.	N/A	N/A
6793674	12/2003	Zapata	N/A	N/A
6858033	12/2004	Kobayashi	N/A	N/A
6921405	12/2004	Feingold et al.	N/A	N/A
6923815	12/2004	Brady et al.	N/A	N/A
6976989	12/2004	Vincent	N/A	N/A
7014641	12/2005	Kobayashi et al.	N/A	N/A
7025782	12/2005	Kobayashi et al.	N/A	N/A
7033366	12/2005	Brady	N/A	N/A
7037312	12/2005	Kikuchi et al.	N/A	N/A
7074227	12/2005	Portney	N/A	N/A
7097649	12/2005	Meyer	N/A	N/A
7131976	12/2005	Kobayashi et al.	N/A	N/A
7156854	12/2006	Brown et al.	N/A	N/A
7348038	12/2007	Makker et al.	N/A	N/A

7422604	12/2007	Vaquero et al.	N/A	N/A
7429263	12/2007	Vaquero et al.	N/A	N/A
7458976	12/2007	Peterson et al.	N/A	N/A
7476230	12/2008	Ohno et al.	N/A	N/A
7494505	12/2008	Kappelhof et al.	N/A	N/A
7645300	12/2009	Tsai	N/A	N/A
8273122	12/2011	Anderson	N/A	N/A
8382769	12/2012	Inoue	N/A	N/A
8460311	12/2012	Ishii	N/A	N/A
8470032	12/2012	Inoue et al.	N/A	N/A
8475526	12/2012	Pynson	N/A	N/A
8475528	12/2012	Ichinohe et al.	N/A	N/A
8523877	12/2012	Ichinohe et al.	N/A	N/A
8523941	12/2012	Ichinohe et al.	N/A	N/A
8535375	12/2012	Ichinohe et al.	N/A	N/A
8545512	12/2012	Ichinohe et al.	N/A	N/A
8574239	12/2012	Ichinohe et al.	N/A	N/A
8603103	12/2012	Kudo et al.	N/A	N/A
8647382	12/2013	Kudo et al.	N/A	N/A
8702795	12/2013	Shoji et al.	N/A	N/A
8747465	12/2013	Someya et al.	N/A	N/A
8968328	12/2014	Ichinohe et al.	N/A	N/A
9114006	12/2014	Inoue	N/A	N/A
9114007	12/2014	Ichinohe et al.	N/A	N/A
9186246	12/2014	Inoue	N/A	N/A
9220593	12/2014	Ichinohe	N/A	N/A
9237947	12/2015	Valle	N/A	N/A
9289288	12/2015	Someya et al.	N/A	N/A
9314373	12/2015	Kudo et al.	N/A	N/A
9326847	12/2015	Sanger	N/A	N/A
9364320	12/2015	Ichinohe et al.	N/A	N/A
9554894	12/2016	Inoue	N/A	N/A
9572710	12/2016	Kudo et al.	N/A	N/A
9655718	12/2016	Kudo	N/A	N/A
9687340	12/2016	Anderson	N/A	N/A
9877826	12/2017	Kudo et al.	N/A	N/A
9901442	12/2017	Kudo et al.	N/A	N/A
9907647	12/2017	Inoue	N/A	N/A
9980811	12/2017	Kudo et al.	N/A	N/A
10039668	12/2017	Kudo et al.	N/A	N/A
10231826	12/2018	Hangya et al.	N/A	N/A
10383723	12/2018	Kudo	N/A	N/A
10390940	12/2018	Someya et al.	N/A	N/A
10405971	12/2018	Someya et al.	N/A	N/A
10517717	12/2018	Inoue	N/A	N/A
10799339	12/2019	Kudo et al.	N/A	N/A
10849738	12/2019	Kudo et al.	N/A	N/A
11033382	12/2020	Watanabe et al.	N/A	N/A
11439499	12/2021	Wensrich et al.	N/A	N/A
11938019	12/2023	Someya et al.	N/A	N/A

12076231	12/2023	Noda et al.	N/A	N/A
2001/0007942	12/2000	Kikuchi et al.	N/A	N/A
2001/0020171	12/2000	Heyman	N/A	N/A
2002/0103490	12/2001	Brady	N/A	N/A
2002/0151904	12/2001	Feingold et al.	N/A	N/A
2002/0165610	12/2001	Waldock	N/A	N/A
2002/0193805	12/2001	Ott et al.	N/A	N/A
2003/0036765	12/2002	Van Noy	N/A	N/A
2003/0040755	12/2002	Meyer	N/A	N/A
2003/0050647	12/2002	Brady	N/A	N/A
2003/0088253	12/2002	Seil	N/A	N/A
2003/0139749	12/2002	Kikuchi et al.	N/A	N/A
2003/0181921	12/2002	Jeannin	N/A	N/A
2003/0195522	12/2002	McNicholas	N/A	N/A
2003/0212406	12/2002	Kobayashi et al.	N/A	N/A
2003/0212407	12/2002	Kikuchi	N/A	N/A
2003/0212408	12/2002	Kobayashi	N/A	N/A
2003/0212409	12/2002	Kobayashi et al.	N/A	N/A
2004/0039345	12/2003	Benz et al.	N/A	N/A
2004/0111094	12/2003	Meyer	N/A	N/A
2004/0117012	12/2003	Vincent	N/A	N/A
2004/0127911	12/2003	Figueroa	N/A	N/A
2004/0147938	12/2003	Dusek et al.	N/A	N/A
2004/0186428	12/2003	Ray	N/A	N/A
2004/0238392	12/2003	Peterson et al.	N/A	N/A
2004/0243141	12/2003	Brown et al.	N/A	N/A
2005/0033308	12/2004	Callahan et al.	N/A	N/A
2005/0049605	12/2004	Vaquero et al.	N/A	N/A
2005/0049606	12/2004	Vaquero et al.	N/A	N/A
2005/0055011	12/2004	Enggaard	N/A	N/A
2005/0125000	12/2004	Tourrette et al.	N/A	N/A
2005/0143750	12/2004	Vaquero	N/A	N/A
2005/0182419	12/2004	Tsai	N/A	N/A
2005/0222578	12/2004	Vaquero	N/A	N/A
2005/0261703	12/2004	Feingold et al.	N/A	N/A
2006/0085013	12/2005	Dusek	N/A	N/A
2006/0142781	12/2005	Pynson et al.	N/A	N/A
2006/0167466	12/2005	Dusek	N/A	N/A
2006/0173540	12/2005	Vincent	N/A	N/A
2006/0200167	12/2005	Peterson et al.	N/A	N/A
2006/0229633	12/2005	Shepherd	N/A	N/A
2006/0235429	12/2005	Lee et al.	N/A	N/A
2006/0293694	12/2005	Futamura	N/A	N/A
2007/0000801	12/2006	Mauran et al.	N/A	N/A
2007/0005135	12/2006	Makker et al.	N/A	N/A
2007/0168026	12/2006	Nagasaka	N/A	N/A
2007/0173860	12/2006	Iwasaki	N/A	N/A
2007/0270945	12/2006	Kobayashi	N/A	N/A
2008/0027460	12/2007	Kobayashi	N/A	N/A
2008/0033449	12/2007	Cole et al.	N/A	N/A

2008/0058830	12/2007	Cole et al.	N/A	N/A
2008/0086146	12/2007	Ishii et al.	N/A	N/A
2008/0097459	12/2007	Kammerlander et al.	N/A	N/A
2008/0221584	12/2007	Downer	N/A	N/A
2009/0036898	12/2008	Ichinohe	N/A	N/A
2009/0043313	12/2008	Ichinohe	N/A	N/A
2009/0112223	12/2008	Downer et al.	N/A	N/A
2009/0125034	12/2008	Pynson	N/A	N/A
2009/0138022	12/2008	Tu et al.	N/A	N/A
2009/0204122	12/2008	Ichinohe et al.	N/A	N/A
2009/0216244	12/2008	Pynson	N/A	N/A
2009/0248031	12/2008	Ichinohe	N/A	N/A
2009/0270876	12/2008	Hoffmann et al.	N/A	N/A
2009/0292293	12/2008	Bogaert et al.	N/A	N/A
2010/0082037	12/2009	Kobayashi et al.	N/A	N/A
2010/0094309	12/2009	Boukhny et al.	N/A	N/A
2010/0106160	12/2009	Tsai	N/A	N/A
2010/0161049	12/2009	Inoue	N/A	N/A
2010/0185206	12/2009	Ichinohe et al.	N/A	N/A
2010/0217273	12/2009	Someya et al.	N/A	N/A
2010/0286704	12/2009	Ichinohe et al.	N/A	N/A
2010/0331808	12/2009	Py et al.	N/A	N/A
2011/0046633	12/2010	Pankin et al.	N/A	N/A
2011/0046635	12/2010	Pankin et al.	N/A	N/A
2011/0082463	12/2010	Inoue	N/A	N/A
2011/0098717	12/2010	Inoue	N/A	N/A
2011/0144654	12/2010	Isaacs et al.	N/A	N/A
2011/0172676	12/2010	Chen	N/A	N/A
2011/0264101	12/2010	Inoue et al.	N/A	N/A
2011/0270264	12/2010	Shoji et al.	N/A	N/A
2011/0288557	12/2010	Kudo et al.	N/A	N/A
2012/0022548	12/2011	Zacharias	N/A	N/A
2012/0022549	12/2011	Someya et al.	N/A	N/A
2012/0071887	12/2011	Ichinohe et al.	N/A	N/A
2012/0123438	12/2011	Horvath et al.	N/A	N/A
2012/0221102	12/2011	Tanaka et al.	N/A	N/A
2013/0006259	12/2012	Sanger	N/A	N/A
2013/0018460	12/2012	Anderson	N/A	N/A
2013/0085507	12/2012	Nagasaka	N/A	N/A
2013/0226193	12/2012	Kudo et al.	N/A	N/A
2013/0245635	12/2012	Inoue	N/A	N/A
2013/0345713	12/2012	Cole et al.	N/A	N/A
2014/0081284	12/2013	Ichinohe et al.	N/A	N/A
2014/0107660	12/2013	Ichinohe et al.	N/A	N/A
2014/0114323	12/2013	Kudo et al.	N/A	N/A
2014/0135784	12/2013	Maroscheck	N/A	N/A
2014/0180299	12/2013	Ichinohe et al.	N/A	N/A
2014/0180300	12/2013	Ichinohe et al.	N/A	N/A
2014/0194890	12/2013	Kudo et al.	N/A	N/A

2014/0200588	12/2013	Anderson	606/107	A61F 2/1672
2014/0276901	12/2013	Auld	N/A	N/A
2014/0296863	12/2013	Anderson et al.	N/A	N/A
2015/0157500	12/2014	Midorikawa	606/107	A61F 9/00736
2015/0327992	12/2014	Wagner et al.	N/A	N/A
2016/0000556	12/2015	Perera	N/A	N/A
2016/0058554	12/2015	Anderson et al.	N/A	N/A
2016/0113759	12/2015	Inoue	N/A	N/A
2016/0151150	12/2015	Sato	N/A	N/A
2016/0193038	12/2015	Kudo et al.	N/A	N/A
2016/0256316	12/2015	Van Noy	N/A	N/A
2016/0270907	12/2015	Attinger	N/A	N/A
2016/0331587	12/2015	Yamada et al.	N/A	N/A
2016/0346077	12/2015	Someya et al.	N/A	N/A
2017/0079772	12/2016	Kudo	N/A	N/A
2017/0119522	12/2016	Auld	N/A	A61F 2/1672
2017/0151056	12/2016	Inoue	N/A	N/A
2017/0202662	12/2016	Someya et al.	N/A	N/A
2017/0252149	12/2016	Kudo et al.	N/A	N/A
2017/0252150	12/2016	Kudo et al.	N/A	N/A
2017/0258582	12/2016	Kudo et al.	N/A	N/A
2017/0354493	12/2016	Andersen et al.	N/A	N/A
2018/0014996	12/2017	Asbaghi	N/A	N/A
2018/0200046	12/2017	Brown et al.	N/A	N/A
2018/0250125	12/2017	Kudo et al.	N/A	N/A
2018/0353287	12/2017	Kudo et al.	N/A	N/A
2019/0151078	12/2018	Watanabe et al.	N/A	N/A
2020/0113674	12/2019	Someya et al.	N/A	N/A
2021/0145570	12/2020	Kudo	N/A	N/A
2021/0161653	12/2020	Noda et al.	N/A	N/A
2022/0151767	12/2021	Kudo	N/A	N/A
2023/0301833	12/2022	Wu	N/A	N/A
2023/0372083	12/2022	Kudo	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101073519	12/2006	CN	N/A
204601363	12/2014	CN	N/A
3610925	12/1986	DE	N/A
4110278	12/1991	DE	N/A
19544119	12/1996	DE	N/A
20219445	12/2002	DE	N/A
0363213	12/1989	EP	N/A
0727966	12/2002	EP	N/A
1360947	12/2002	EP	N/A
1502559	12/2004	EP	N/A

1790317	12/2006	EP	N/A
1808150	12/2006	EP	N/A
1832247	12/2006	EP	N/A
1338254	12/2007	EP	N/A
2074961	12/2008	EP	N/A
2255751	12/2009	EP	N/A
2286763	12/2010	EP	N/A
2286764	12/2010	EP	N/A
2368526	12/2010	EP	N/A
2574308	12/2012	EP	N/A
2853236	12/2014	EP	N/A
3391855	12/2017	EP	N/A
2749752	12/1996	FR	N/A
63-197453	12/1987	JP	N/A
4-212350	12/1991	JP	N/A
5-103803	12/1992	JP	N/A
5-103808	12/1992	JP	N/A
5-103809	12/1992	JP	N/A
08-019558	12/1995	JP	N/A
8-024282	12/1995	JP	N/A
8-505540	12/1995	JP	N/A
9-506285	12/1996	JP	N/A
11-113939	12/1998	JP	N/A
11-506357	12/1998	JP	N/A
2000-516487	12/1999	JP	N/A
2000-516488	12/1999	JP	N/A
2001-502563	12/2000	JP	N/A
2001-104347	12/2000	JP	N/A
2001-259033	12/2000	JP	N/A
2002-516709	12/2001	JP	N/A
2002-355268	12/2001	JP	N/A
2002-541912	12/2001	JP	N/A
2003-144480	12/2002	JP	N/A
3412106	12/2002	JP	N/A
2003-210498	12/2002	JP	N/A
2003-527162	12/2002	JP	N/A
2003-325569	12/2002	JP	N/A
2003-325570	12/2002	JP	N/A
2003-325572	12/2002	JP	N/A
2004-024854	12/2003	JP	N/A
2004-041271	12/2003	JP	N/A
2004-188194	12/2003	JP	N/A
2004-344213	12/2003	JP	N/A
2004-351196	12/2003	JP	N/A
2006-014963	12/2005	JP	N/A
2006-181269	12/2005	JP	N/A
2006-297146	12/2005	JP	N/A
2006-333924	12/2005	JP	N/A
2006-333980	12/2005	JP	N/A
2006-333981	12/2005	JP	N/A

2007-503872	12/2006	JP	N/A
2007-152010	12/2006	JP	N/A
2007-181604	12/2006	JP	N/A
2007-222309	12/2006	JP	N/A
2007-244570	12/2006	JP	N/A
2007-526091	12/2006	JP	N/A
2007-307168	12/2006	JP	N/A
2008-012016	12/2007	JP	N/A
2008-521535	12/2007	JP	N/A
2008-212689	12/2007	JP	N/A
2008-237274	12/2007	JP	N/A
2009-028223	12/2008	JP	N/A
2009-028233	12/2008	JP	N/A
2009-072221	12/2008	JP	N/A
2011-019987	12/2010	JP	N/A
2011-087976	12/2010	JP	N/A
2011-160858	12/2010	JP	N/A
2011-160859	12/2010	JP	N/A
2013-144163	12/2012	JP	N/A
2014-050484	12/2013	JP	N/A
2014-079630	12/2013	JP	N/A
2016-137122	12/2015	JP	N/A
WO9407436	12/1993	WO	N/A
WO9513022	12/1994	WO	N/A
WO9628122	12/1995	WO	N/A
WO9715253	12/1996	WO	N/A
WO9812969	12/1997	WO	N/A
WO9937247	12/1998	WO	N/A
WO9958086	12/1998	WO	N/A
WO9959668	12/1998	WO	N/A
WO0045746	12/1999	WO	N/A
WO0062712	12/1999	WO	N/A
WO2002071982	12/2001	WO	N/A
WO2002096322	12/2001	WO	N/A
WO2004/041323	12/2003	WO	N/A
WO2004105649	12/2003	WO	N/A
WO2005023154	12/2004	WO	N/A
WO2005030097	12/2004	WO	N/A
WO2005070341	12/2004	WO	N/A
WO2005084588	12/2004	WO	N/A
WO2006070628	12/2005	WO	N/A
WO2006080191	12/2005	WO	N/A
WO2006090531	12/2005	WO	N/A
WO2007037223	12/2006	WO	N/A
WO2007097221	12/2006	WO	N/A
WO2007080869	12/2006	WO	N/A
WO2008149794	12/2007	WO	N/A
WO2008149795	12/2007	WO	N/A
WO2009058929	12/2008	WO	N/A
WO2009148091	12/2008	WO	N/A

WO2010028873	12/2009	WO	N/A
WO2010064970	12/2009	WO	N/A
WO2011126144	12/2010	WO	N/A
WO2011155636	12/2010	WO	N/A
WO2012086797	12/2011	WO	N/A
WO2012155887	12/2011	WO	N/A
WO2015012312	12/2014	WO	N/A
WO2016191764	12/2015	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 17/055/253, filed Nov. 13, 2020. cited by applicant

U.S. Appl. No. 17/055,186, filed Nov. 13, 2020. cited by applicant

U.S. Appl. No. 14/033,888, filed Sep. 23, 2013, U.S. Pat. No. 9,200,593. cited by applicant

U.S. Appl. No. 16/550,144, filed Aug. 23, 2019. cited by applicant

U.S. Appl. No. 16/313,184, filed Dec. 26, 2018. cited by applicant

PCT Written Opinion dated Oct. 3, 2017 for PCT App. Ser. No. PCT/JP2017/023748. cited by applicant

EPO Extended European Search Report dated Feb. 12, 2020 for EPO App. Ser. No. 17820207.3. cited by applicant

U.S. Appl. No. 17/435,762, filed Sep. 2, 2021. cited by applicant

Chinese Office Action dated Oct. 11, 2021 for CN App. Ser. No. 201780040285.3. cited by applicant

U.S. Appl. No. 18/186,167, filed Mar. 18, 2023. cited by applicant

U.S. Appl. No. 17/435,762, filed Sep. 2, 2021, US 20220151757A1. cited by applicant

U.S. Appl. No. 18/044,235, filed Mar. 7, 2023. cited by applicant

Office Action dated Jun. 6, 2023 for Brazilian App. Ser. No. BR112018077090-6, with English language translation. cited by applicant

Jul. 19, 2020. cited by applicant

U.S. Appl. No. 17/801,364, filed Aug. 22, 2022, US 20230033115A1. cited by applicant

U.S. Appl. No. 18/044,235, filed Mar. 7, 2023, US 20230372083A1. cited by applicant

U.S. Appl. No. 18/694,985, filed Mar. 24, 2023. cited by applicant

U.S. Appl. No. 16/550,144, filed Aug. 23, 2019, U.S. Pat. No. 11,617,643. cited by applicant

U.S. Appl. No. 18/186,167, filed Mar. 18, 2023, U.S. Pat. No. 11,938,019. cited by applicant

U.S. Appl. No. 17/055,253, filed Nov. 13, 2020, U.S. Pat. No. 12,076,231. cited by applicant

U.S. Appl. No. 12/602,442, filed Dec. 15, 2009, U.S. Pat. No. 8,747,465. cited by applicant

U.S. Appl. No. 13/244,449, filed Sep. 24, 2011, U.S. Pat. No. 9,289,288. cited by applicant

U.S. Appl. No. 15/063,395, filed Mar. 7, 2016, US 20160346077A1. cited by applicant

U.S. Appl. No. 15/476,717, filed Mar. 31, 2017, US 20170202662A1. cited by applicant

U.S. Appl. No. 12/602,454, filed Dec. 15, 2009, U.S. Pat. No. 8,475,528. cited by applicant

U.S. Appl. No. 13/244,452, filed Sep. 24, 2011, U.S. Pat. No. 8,535,375. cited by applicant

U.S. Appl. No. 12/667,510, filed Dec. 31, 2009, U.S. Pat. No. 9,114,006. cited by applicant

U.S. Appl. No. 14/812,104, filed Jul. 29, 2015, U.S. Pat. No. 9,907,647. cited by applicant

U.S. Appl. No. 12/995,263, filed Dec. 15, 2010, U.S. Pat. No. 9,554,894. cited by applicant

U.S. Appl. No. 15/382,377, filed Dec. 16, 2016, US 20170151056A1. cited by applicant

U.S. Appl. No. 12/997,651, filed Dec. 13, 2010, U.S. Pat. No. 8,382,769. cited by applicant

U.S. Appl. No. 13/757,790, filed Feb. 2, 2012, U.S. Pat. No. 9,186,246. cited by applicant

U.S. Appl. No. 13/583,216, filed Apr. 6, 2011, U.S. Pat. No. 9,326,847. cited by applicant

U.S. Appl. No. 13/699,708, filed Jun. 8, 2011, U.S. Pat. No. 8,647,382. cited by applicant

U.S. Appl. No. 14/145,846, filed Dec. 31, 2013, U.S. Pat. No. 9,314,373. cited by applicant

U.S. Appl. No. 15/071,880, filed Mar. 16, 2016, U.S. Pat. No. 10,039,668. cited by applicant

U.S. Appl. No. 15/336,678, filed Oct. 27, 2016, U.S. Pat. No. 9,572,710. cited by applicant
U.S. Appl. No. 15/608,895, filed May 30, 2017, U.S. Pat. No. 9,980,811. cited by applicant
U.S. Appl. No. 13/059,401, filed Feb. 16, 2011, U.S. Pat. No. 8,702,795. cited by applicant
U.S. Appl. No. 13/061,143, filed Feb. 26, 2011, U.S. Pat. No. 8,470,032. cited by applicant
U.S. Appl. No. 13/143,322, filed Jul. 5, 2011, U.S. Pat. No. 8,603,103. cited by applicant
U.S. Appl. No. 14/099,989, filed Dec. 8, 2013, U.S. Pat. No. 9,655,718. cited by applicant
U.S. Appl. No. 15/600,679, filed May 19, 2017, U.S. Pat. No. 9,877,826. cited by applicant
U.S. Appl. No. 15/600,684, filed May 19, 2017, U.S. Pat. No. 9,901,442. cited by applicant
U.S. Appl. No. 11/814,508, filed Jul. 23, 2007, U.S. Pat. No. 8,545,512. cited by applicant
U.S. Appl. No. 14/033,888, filed Sep. 23, 2013, U.S. Pat. No. 9,220,593. cited by applicant
U.S. Appl. No. 11/816,676, filed Aug. 20, 2007, U.S. Pat. No. 8,523,877. cited by applicant
U.S. Appl. No. 13/966,209, filed Aug. 13, 2013, U.S. Pat. No. 9,364,320. cited by applicant
U.S. Appl. No. 12/095,172, filed May 28, 2008, U.S. Pat. No. 8,523,941. cited by applicant
U.S. Appl. No. 14/011,018, filed Aug. 27, 2013, U.S. Pat. No. 8,968,328. cited by applicant
U.S. Appl. No. 12/088,328, filed Mar. 27, 2008, U.S. Pat. No. 8,574,239. cited by applicant
U.S. Appl. No. 14/065,365, filed Oct. 28, 2013, U.S. Pat. No. 9,114,007. cited by applicant
U.S. Appl. No. 11/722,601, filed Apr. 10, 2008, U.S. Pat. No. 8,460,311. cited by applicant
U.S. Appl. No. 15/126,277, filed Sep. 14, 2016, US 20170079772A1. cited by applicant
U.S. Appl. No. 15/756,565, filed Feb. 28, 2018, US 20180250125A1. cited by applicant
U.S. Appl. No. 15/756,569, filed Feb. 28, 2018, US 20180353287A1. cited by applicant
U.S. Appl. No. 16/313,180, filed Dec. 26, 2018. cited by applicant
U.S. Appl. No. 16/313,184, filed Dec. 26, 2018, US 20190151078A1. cited by applicant
U.S. Appl. No. 15/063,395, filed Mar. 7, 2016, U.S. Pat. No. 10,390,940. cited by applicant
U.S. Appl. No. 15/476,717, filed Mar. 31, 2017, U.S. Pat. No. 10,405,971. cited by applicant
U.S. Appl. No. 16/550,144, filed Aug. 23, 2019, US 20200113674A1. cited by applicant
U.S. Appl. No. 15/382,377, filed Dec. 16, 2016, U.S. Pat. No. 10,517,717. cited by applicant
U.S. Appl. No. 15/126,277, filed Sep. 14, 2016, U.S. Pat. No. 10,383,723. cited by applicant
U.S. Appl. No. 15/756,565, filed Feb. 28, 2018, U.S. Pat. No. 10,849,738. cited by applicant
U.S. Appl. No. 15/756,569, filed Feb. 28, 2018, U.S. Pat. No. 10,799,339. cited by applicant
U.S. Appl. No. 16/313,180, filed Dec. 26, 2018, US 20190192284A1. cited by applicant
U.S. Appl. No. 16/313,184, filed Dec. 26, 2018, U.S. Pat. No. 11,033,382 cited by applicant
U.S. Appl. No. 17/055,253, filed Nov. 13, 2020, US 20210161653A1. cited by applicant
U.S. Appl. No. 17/055,186, filed Nov. 13, 2020, US 20210145570A1. cited by applicant
U.S. Appl. No. 17/435,762, filed Sep. 2, 2021, US 20220151767A1. cited by applicant
U.S. Appl. No. 17/801,364, filed Aug. 22, 2022. cited by applicant
Presentation given by James P. McCulley, titled “Benefits of Newest Generation Fully Preloaded Aspheric IOL Delivery System,” on Sep. 16, 2008 at the “Aspheric IOLs” free paper session of the 2008 Congress of the European Society of Cataract and Refractive Surgery, in Berlin, Germany. cited by applicant

Primary Examiner: Dang; Anh T

Attorney, Agent or Firm: Henricks Slavin LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an intraocular lens injector used for injecting an intraocular lens

into an eyeball (intraocular).

DESCRIPTION OF RELATED ART

(2) Cataract surgery involves the removal of a cloudy crystalline lens by ultrasonic emulsification followed by implantation on an intraocular lens into an eye. Currently, an intraocular lens made of a soft material such as silicone elastomer is used and is injected into an eye using an intraocular lens injector. Patent document 1 discloses an intraocular lens injector with the following configuration: an injector main body containing an injection tube which in turn contains a nozzle portion. At the time of an actual surgery, the nozzle portion of the injection tube is injected into an incisional wound of an eyeball, and the intraocular lens is released from the nozzle portion by a rod contained within the intraocular lens injector.

(3) Meanwhile, patent document 2 discloses a configuration of a cartridge used for injecting the intraocular lens, in which a circumferential protrusion is formed on an outer surface of the nozzle portion. When the nozzle portion is inserted into the incisional wound of the eyeball, the circumferential protrusion is brought into contact with the surface of the eyeball, thereby preventing the nozzle portion from being completely inserted into the incisional wound.

PRIOR ART DOCUMENT

Patent Document

(4) [Patent Document 1] Japanese Unexamined Patent Application Publication No. 2015-177845

(5) [Patent Document 2] Japanese Patent No. 4685897

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) Regarding the above-described technologies described in patent documents 1 and 2, each has a different merit.

(7) Namely, in the technique described in patent document 1, the nozzle portion can be inserted deeply into the incisional wound of the eyeball since there is no circumferential protrusion on the outer surface of the nozzle portion. When the nozzle portion is inserted deeply into the incisional wound of the eyeball, the intraocular lens can be guided into a lens capsule in a more reliable manner as compared to a shallow insertion. Accordingly, it is possible to forgo the trouble of using forceps or the like for housing the intraocular lens in the lens capsule, the intraocular lens being injected into the eye by the intraocular lens injector.

(8) In the technique described in patent document 2, the nozzle portion can be inserted shallowly into the incisional wound of the eyeball due to the presence of a circumferential protrusion on the outer surface of the nozzle. In this instance the intraocular lens can be injected through a smaller incisional wound as compared with a case of a deep insertion. Accordingly, it is possible to use a Wound-assisted method, which reduces the size of the incisional wound and improves healing after surgery. The Wound-assisted method makes it possible to reduce the size of the incisional wound necessary for passage of the intraocular lens by using a tunnel portion of the incisional wound as an injection path for the intraocular lens.

(9) Both techniques described above have their own advantages and benefits. Thus it is not surprising that among intraocular lens injector users (principally surgeons such as ophthalmologists), some will prefer a procedure to insert the nozzle portion deeply into the incisional wound of the eyeball while some will prefer the procedure to insert the nozzle portion shallowly, when performing cataract surgery. In order to cater to the preference of each user's technique, it is necessary to prepare both of having a nozzle portion with a circumferential protrusion or the like formed thereon, and an injector having the nozzle portion with no circumferential protrusion formed thereon.

(10) A main object of the present invention is to provide the intraocular lens injector capable of flexibly responding to differences in intraocular lens injection procedure and technique.

Means for Solving the Problem

(11) A first aspect of the present invention is an intraocular lens injector for injecting an intraocular

lens into an eye, including:

(12) a hollow body having a nozzle portion; and

(13) an attachment member having a protruding surface disposed in a state of protruding to the outside from an outer circumferential surface of the nozzle portion,

(14) wherein the attachment member is configured movably with respect to the hollow body.

(15) A second aspect of the present invention is the intraocular lens injector according to the first aspect, wherein the attachment member is detachable and attachable from/to the hollow body.

(16) A third aspect of the present invention is the intraocular lens injector according to the first or second aspect, wherein the attachment member is movable in an axial direction of the hollow body.

(17) A fourth aspect of the present invention is the intraocular lens according to the third aspect, wherein the attachment member is movable in the axial direction of the hollow body while being attached to the hollow body.

(18) A fifth aspect of the present invention is the intraocular lens according to any one of the first to fourth aspects, wherein an engaging portion which is engaged with the hollow body is formed in the attachment member, and an engaged portion is formed in the hollow body corresponding to the engaging portion.

(19) A sixth aspect of the present invention is the intraocular lens injector according to the fifth aspect, wherein the engaging portion has a lock mechanism in which the attachment member is temporarily fixed to the hollow body by an engagement between the engaging portion and the engaged portion.

(20) A seventh aspect of the present invention is the intraocular lens injector according to the fifth aspect, wherein the engaging portion includes a tongue piece having a plurality of engagement holes formed thereon, and the engaged portion includes a locking claw engaged with one of the plurality of engagement holes.

(21) An eighth aspect of the present invention is the intraocular lens injector according to any one of the first to seventh aspects, wherein the attachment member is configured to restrict an injection amount by the protruding surface, when the nozzle portion is inserted into an incisional wound of an eyeball.

(22) A ninth aspect of the present invention is the intraocular lens injector according to any one of the first to eighth aspects, wherein a viewing window is provided on the attachment member.

(23) A tenth aspect of the present invention is the intraocular lens injector according to the ninth aspect, wherein the protruding surface is formed in a shape with an upper part cut out, and the viewing window is formed so as to communicate with a cutout portion of the protruding surface.

(24) An eleventh aspect of the present invention is the intraocular lens injector according to any one of the first to tenth aspects, wherein the protruding surface is made of a soft resin.

(25) A twelfth aspect of the present invention is the intraocular lens injector according to any one of the first to eleventh aspects, wherein the nozzle portion has a cutout portion formed along an axial direction of the hollow body, and a tapered portion is formed on the attachment member, to reduce an outer circumferential diameter of the nozzle portion, by being brought into contact with an outer circumferential surface of the nozzle portion, which is a portion where the cutout portion is formed.

(26) A thirteenth aspect of the present invention is the intraocular lens injector according to any one of the first to twelfth aspects, wherein the hollow body has a lens installing portion, and an intraocular lens is installed on the lens installing portion.

Advantage of the Invention

(27) According to the present invention, it is possible to respond flexibly to differences in procedures of intraocular lens injection surgery.

Description

BRIEF DESCRIPTION OF THE DRAWING

- (1) FIG. 1 is a perspective view showing a configuration example of an intraocular lens injector according to a first embodiment of the present invention.
- (2) FIG. 2A is a plan view showing a configuration example of the intraocular lens injector according to the first embodiment of the present invention, FIG. 2B is a side view, FIG. 2C is a sectional view taken along the line E1-E1.
- (3) FIG. 3 is a perspective view showing a configuration of an injector main body.
- (4) FIG. 4A is a side view showing a configuration of an injector main body, FIG. 4B is a sectional view taken along the line F1-F1 in FIG. 4A, and FIG. 4C is an enlarged view of a portion F2 in FIG. 4B.
- (5) FIG. 5A is a perspective view showing an injector main body in a state in which a rotation restricting part is opened, and FIG. 5B is a view seen from arrow G in FIG. 5A.
- (6) FIG. 6 is a perspective view showing a configuration and an arrangement of a tip part of the injector main body.
- (7) FIGS. 7A to 7C show the configuration of an injection tube, wherein FIG. 7A is a plan view, FIG. 7B is a side view, and FIG. 7C is a bottom view.
- (8) FIG. 8 is a cross-sectional view taken along the line H-H in FIG. 7B.
- (9) FIG. 9 is a perspective view showing a configuration of an attachment member.
- (10) FIGS. 10A to 10C show a configuration of an attachment member, wherein FIG. 10A is a plan view, FIG. 10B is a side view, and FIG. 10C is a rear view.
- (11) FIG. 11 shows a state in which an attachment member is attached to the injection tube, wherein FIG. 11A is a plan view, FIG. 11B is a side view, and FIG. 11C is a bottom view.
- (12) FIGS. 12A and 12B show a preferable embodiment when the attachment member is used, wherein FIG. 12A is a cross-sectional view as seen from the axial direction, and FIG. 12B is a cross-sectional view as seen from a direction orthogonal to the axial direction.
- (13) FIG. 13A to FIG. 13C are views for explaining a procedure of attaching and detaching the attachment member, respectively.
- (14) FIG. 14 is an enlarged view of a portion E5 in FIG. 2C.
- (15) FIG. 15 is a cross-sectional view taken along the line E2-E2 in FIG. 2B.
- (16) FIG. 16A is a cross-sectional view taken along line E3-E3 in FIG. 2B, and FIG. 16B is a cross-sectional view taken along the line E4-E4 in FIG. 2B.
- (17) FIG. 17 is a view for explaining a function of the attachment member.
- (18) FIG. 18 is a view showing an example of changing a protrusion amount of the nozzle member.
- (19) FIG. 19 is a perspective view showing a state in which the attachment member is attached to the injection tube in the intraocular lens injector according to a second embodiment of the present invention.
- (20) FIG. 20 is a perspective view showing a configuration of the injector tube according to the second embodiment of the present invention.
- (21) FIG. 21 is a perspective view showing a configuration of the attachment member according to the second embodiment of the present invention.
- (22) FIGS. 22A to 22D show the configuration of the attachment member according to the second embodiment of the present invention, wherein FIG. 22A is a plan view, FIG. 22B is a side view, FIG. 22C is a bottom view, and FIG. 22D is a rear view.
- (23) FIGS. 23A to 23C show a state in which the attachment member is attached to the injection tube, wherein FIG. 23A is a plan view, FIG. 23B is a side view, and FIG. 23C is a bottom view.
- (24) FIG. 24A is a cross-sectional view taken along the line J2-J2 in FIG. 23B, FIG. 24B is a cross-sectional view taken along the line J1-J1 in FIG. 23A, and FIG. 24C is an enlarged view of portion K in FIG. 24B.
- (25) FIG. 25A is a side view showing a state in which a locking claw is engaged with a rear

engagement hole, FIG. 25B is a side view showing a state in which the locking claw is engaged with a central engagement hole, and FIG. 25C is a side view showing a state in which the locking claw is engaged with a front engagement hole.

(26) FIG. 26 is a perspective view showing a configuration of the intraocular lens injector according to a third embodiment of the present invention.

(27) FIG. 27A is a plan view showing a configuration of the intraocular lens injector according to a third embodiment of the present invention, wherein FIG. 27B is a side view, and FIG. 27C is a cross-sectional view taken along the line M-M in FIG. 27A.

(28) FIGS. 28A and 28B are perspective views showing a configuration of the injector main body according to the third embodiment of the present invention, FIG. 28A shows a state in which the rotation restricting portion provided in the injector main body is opened, and FIG. 28B shows a state in which the rotation restricting portion is closed.

(29) FIGS. 29A to 29C show a configuration of the injection tube according to a third embodiment of the present invention, wherein FIG. 29A is a plan view, FIG. 29B is a side view, and FIG. 29C is a bottom view.

(30) FIG. 30A is a perspective view of the injection tube as seen obliquely from above according to the third embodiment of the present invention, and FIG. 30B is a perspective view as seen diagonally from below.

(31) FIGS. 31A and 31B are perspective views of the attachment member according to the third embodiment of the present invention as seen obliquely from above, wherein FIG. 31A shows the attachment member before being attached to the injection tube, and FIG. 31B shows the attachment member after being attached to the injection tube.

(32) FIGS. 32A to 32C show a configuration of the attachment member according to the third embodiment of the present invention, wherein FIG. 32A is a plan view, FIG. 32B is a side view, and FIG. 32C is a bottom view.

(33) FIG. 33 is a view as seen from arrow N in FIG. 32B.

(34) FIG. 34 is a longitudinal sectional view of the attachment member according to the third embodiment of the present invention, showing a state in which the engaging portion is closed.

(35) FIG. 35 is a perspective view (part 1) showing a procedure for attaching the attachment member according to the third embodiment of the present invention.

(36) FIG. 36 is a perspective view (part 2) showing a procedure for attaching the attachment member according to the third embodiment of the present invention.

(37) FIG. 37 is a longitudinal sectional view showing a state in which the attachment member is attached to the injection tube according to the third embodiment of the present invention.

(38) FIG. 38 is a longitudinal sectional view showing a state in which the attachment member attached to the injection tube is moved in an axial direction according to the third embodiment of the present invention.

(39) FIGS. 39A to 39D are section views showing an arrangement relationship between a plunger and a rotation restricting portion according to the third embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(40) Embodiments of the present invention will be described hereafter, with reference to the drawings.

(41) <Configuration of the Intraocular Lens Injector>

(42) FIG. 1 is a perspective view showing a configuration example of an intraocular lens injector according to a first embodiment of the present invention. Further, FIG. 2A is a plan view showing a configuration example of the intraocular lens injector according to the first embodiment of the present invention, FIG. 2B is a side view, FIG. 2C is a cross-sectional view taken along the line E1-E1.

(43) An intraocular lens injector 1 is used for injecting an intraocular lens into an eye. In this embodiment, as an example of the intraocular lens, there is provided an intraocular lens 4 as a one-

piece type intraocular lens **4** (see FIG. 6) made of a soft material such as silicone elastomer or soft acrylic, including a circular optical portion **4a** which performs an optical function and two supporting portions **4b** that curve outward from two positions on the outer circumferential portion of the optical portion **4a** and extend outward.

(44) Further, in this embodiment, in describing a relative positional relationship and a direction of movement and the like of each part of the intraocular lens injector **1**, one of the X axis directions is defined as X1 direction, the other direction is defined as X2 direction, and one of the Y axis directions is defined as Y1 direction, the other direction is defined as Y2 direction, and one of the Z axis directions is defined as Z1 direction and the other direction is defined as Z2 direction, and X1 direction is defined as a front side (frontward), X2 direction is defined as a rear end side (rearward), Y1 direction is defined as a left side (leftward), and Y2 direction is defined as a right side (rightward), Z1 direction is defined as an upper side (upward), and Z2 direction is defined as a downside (downward). Among them, the X axis direction (X1 direction and X2 direction) corresponds to a length direction of the intraocular lens injector **1**, and the Y axis direction (Y1 direction and Y2 direction) corresponds to a width direction of the intraocular lens injector **1**, and the Z axis direction (Z1 direction and Z2 direction) corresponds to a height direction of the intraocular lens injector **1**.

(45) The intraocular lens injector **1** has a configuration including an injector main body **5**, a slider **6**, an injection tube **7**, a rotary member **8**, a plunger **9**, a rod **10** (see FIG. 2C), and an attachment member **16**. These constituent elements are preferably constituted by resin molded products, respectively. The injector main body **5** and the injection tube **7** have a hollow structure and are coupled to each other to thereby constitute a hollow body. The slider **6** is attached to the injector main body **5**. The injection tube **7** is coupled to the tip part of the injector main body **5**. The rotary member **8** is rotatably connected to a rear end portion of the injector main body **5**. The plunger **9** is disposed coaxially with the injector main body **5**. A part of the plunger **9** is disposed inside of the injector main body **5** through the rotary member **8**, and the other part of the plunger **9** is disposed to protrude rearward from the rotary member **8**. The rod **10** is disposed inside of the hollow body which is composed of the injector main body **5** and the injection tube **7**. The attachment member **16** is a member attached to the hollow body so as to provide a predetermined additional function.

(46) (Injector Main Body)

(47) FIG. 3 is a perspective view showing the configuration of the injector main body. Further, FIG. 4A is a side view showing a configuration of an injector main body, FIG. 4B is a cross-sectional view taken along the line F1-F1 in FIG. 4A, and FIG. 4C is an enlarged view of a portion F2 in FIG. 4B.

(48) The injector main body **5** is formed in a cylindrical shape as a whole. A lens installing portion **11** is provided at the tip part of the injector main body **5**. The intraocular lens **4** is installed on the lens installing portion **11**. The lens installing portion **11** is formed so as to protrude forward from an outer circumferential wall on a lower side of the injector main body **5**.

(49) An injection tube coupling portion **5a** is formed on the outer circumferential portion on the tip side of the injector main body **5**. Slits **12** are respectively formed on both left and right sides of the injector main body **5**. The slits **12** are formed on the tip side of the injector main body **5**. When the slider **6** is attached to the injector main body **5**, the slits **12** movably support the slider **6** in the axial direction (central axis direction) of the injector main body **5**.

(50) The rear end of the injector main body **5** opens in a circular shape, through which the plunger **9** and the rod **10** can be inserted into the injector main body **5**. A rotation restricting portion **13**, a flange portion **14**, and a rotation supporting portion **15** are formed at the rear end portion of the injector main body **5** and in the vicinity thereof. The rotation restricting portion **13** restricts a rotation of the plunger **9** inserted into the injector main body **5**.

(51) FIGS. 4A to 4C show the injector main body in a state in which the rotation restricting portion is closed. In contrast, FIG. 5A is a perspective view showing the injector main body in a state in

which the rotation restricting portion is opened, and FIG. 5B is a view seen from arrow G in FIG. 5A.

(52) The rotation restricting portion **13** is formed slightly forward of the flange portion **14**. The rotation restricting portion **13** constitutes a part of an outer circumferential wall of the injector main body **5**, and is provided so as to be rotatable around a pivotally supporting portion **13a** so as to open and close an opening formed on the outer circumferential wall. Further, a pair of sliding guides **13b** is formed in the rotation restricting portion **13**, and a recessed guide groove **13c** is formed between these sliding guides **13b**. The pair of sliding guides **13b** is disposed in a tube of the injector main body **5** when the rotation restricting portion **13** is closed, and is disposed outside of the tube of the injector main body **5** when the rotation restricting portion **13** is opened. A closed state of the rotation restricting portion **13** is maintained, for example, by an engagement of recessed and protruded portions (not shown). When the rotation restricting portion **13** is rotated around the pivotally supporting portion **13a** against a holding force caused by the engagement of the recessed and protruded portions, as shown in FIG. 5, the rotation restricting portion **13** is opened. Tip surfaces of the respective sliding guides **13b** are formed as anti-falloff portions **13d** (see FIG. 4C). The anti-falloff portions **13d** prevent the plunger **9** from being pulled out from the injector main body **5**.

(53) The flange portion **14** is a portion for hooking a finger (usually an index finger and a middle finger) of a user's hand, when operating the intraocular lens injector **1** by a push system. The rotation supporting portion **15** is provided slightly rearward of the flange portion **14**. The rotation supporting portion **15** is formed in a ring shape having a stepped structure protruding in a ring shape.

(54) As shown in FIG. 6, the lens installing portion **11** includes a bottom surface portion **11a**, a lens receiving portion **11b**, and a lens guide portion **11c**. The lens receiving portion **11b** receives and supports the intraocular lens **4** from below. The intraocular lens **4** is installed on the lens installing portion **11** in a state in which one supporting portion **4b** is disposed in front and the other supporting portion **4b** is disposed in rear. The intraocular lens injector **1** is of a preload type in which the intraocular lens **4** is previously installed on the lens installing portion **11** of the injector main body **5**. Therefore, the intraocular lens **4** is one of the components of the intraocular lens injector **1**. However, in executing the present invention, the intraocular lens injector **1** is not necessarily required to be the preload type.

(55) A center portion in a width direction of the bottom surface portion **11a** is slightly recessed. The lens receiving portion **11b** is formed on both left and right sides of the lens installing portion **11**. The lens receiving portion **11b** is formed to be one step higher than the bottom surface portion **11a**. This is because when the intraocular lens **4** is supported on the lens receiving portion **11b**, the optical portion **4a** of the intraocular lens **4** is supported in a floating state from the bottom surface portion **11a** without contacting the bottom surface portion **11a**. In the same manner as in the lens receiving portion **11b**, the lens guide portion **11c** is formed on both right and left sides of the lens installing portion **11**. The lens guide portion **11c** guides the optical portion **4a** of the intraocular lens **4** supported by the lens receiving portion **11b** so as to sandwich the optical portion **4a** from both the left and right sides. The lens guide portion **11c** is formed in a vertically upright state from the lens receiving portion **11b**. On upper end portions of the left and right lens guide portions **11c**, an inclined surface **11d** and a restricting portion **11e** are formed adjacent to each other in the axial direction of the injector main body **5**. The inclined surface **11d** is formed so as to incline outwardly to easily receive the intraocular lens **4** on the lens installing portion **11**. The inclined surface **11d** is formed on the tip side of the lens installing portion **11** with respect to the restricting portion **11e** in the axial direction of the injector main body **5**. The restricting portion **11e** is formed to protrude toward the center in the width direction of the lens installing portion **11**. The restricting portion **11e** limits a movable range in a vertical direction of the intraocular lens **4** which is supported by the lens receiving portion **11b**.

(56) (Slider)

(57) As shown in FIG. 1, FIG. 2A, FIG. 2B, and FIG. 6, the slider 6 has a configuration including a lens pressing portion 6a and a pair of wing portions 6b. The lens pressing portion 6a is disposed at the tip part of the slider 6, and the pair of wing portions 6b is disposed in pairs on the left and right sides of the slider 6. When the slider 6 is moved forward, as shown in FIG. 6, the lens pressing portion 6a is disposed so as to advance to the upper side of the optical portion 4a of the intraocular lens 4 which is installed on the lens installing portion 11.

(58) The pair of wing portions 6b is disposed outside of the outer circumferential wall portion of the injector main body 5. The pair of wing portions 6b is the portions with which a user's finger (usually an index finger and thumb) is brought into contact, when the slider 6 is moved in the axial direction of the injector main body 5 during use of the intraocular lens injector 1. The user is a surgeon such as an ophthalmologist performing surgery or a nurse assisting a surgeon. On the outer surface of each wing portion 6b, unevenness for anti-falloff and a mark (triangular arrow in this embodiment) indicating the moving direction of the slider 6 are formed. Further, the outer surface of each wing portion 6b is curved from the rear end side toward the tip side so that the fingers of the user are easily caught, and unevenness for anti-falloff is formed on this curved portion.

(59) (Injection Tube)

(60) FIGS. 7A to 7C show the configuration of an injection tube, wherein FIG. 7A is a plan view, FIG. 7B is a side view, and FIG. 7C is a bottom view. Further, FIG. 8 is a cross-sectional view taken along the line H-H in FIG. 7B.

(61) The injection tube 7 is a tube for folding the intraocular lens 4 small and guiding it into the eye, when the intraocular lens 4 installed on the lens installing portion 11 is injected into the eye. The injection tube 7 is made of a transparent or translucent material, so that a state of the intraocular lens 4 moving inside of the insertion tube 7 can be visually recognized from the outside.

(62) The injection tube 7 has a hollow injection tube main body 7a and a narrow tubular nozzle portion 7b. The injection tube 7 is attached to the tip part of the injection main body 5. At this time, the lens installing portion 11 of the injector main body 5 is housed and disposed inside of the injector main body 7a of the injection tube 7, together with the intraocular lens 4 installed thereon. An injection hole 7c is formed on an upper surface of the injection tube main body 7a. The injection hole 7c is a hole through which a viscoelastic substance (for example, sodium hyaluronate etc.) is injected. The viscoelastic substance injected from the injection hole 7c is supplied to the intraocular lens 4 installed on the lens installing portion 11.

(63) Meanwhile, the rear end portion of the injection tube main body 7a is opened, and hooking portions 7d and wedge portions 7e are formed around the opening portion. The hooking portions 7d are disposed in pairs at the top and bottom, and the wedge portions 7e are disposed in pairs on the right and left. The hooking portions 7d are portions to be hooked on the injection tube coupling portion 5a of the injector main body 5 when the injection tube 7 is attached to the tip part of the injector main body 5. The wedge portions 7e are portions to be inserted into an entrance portion of the slit 12 of the injector main body 5 when the injection tube 7 is attached to the injector main body 5.

(64) The injection tube main body 7a is divided into a first portion 7f and a second portion 7g in the axial direction of the injection tube 7. The first portion 7f has an internal space capable of housing the lens installing portion 11 of the injector main body 5, and is formed wider than the second portion 7g. The second portion 7g is located forward of the first portion 7f. The internal space of the injector main body 7a is gradually narrowed from the first portion 7f to the second portion 7g in order to fold the intraocular lens 4 small when the intraocular lens 4 is pushed out by the rod 10.

(65) A first protrusion 7h, a second protrusion 7i, and a third protrusion 7j are formed on the second portion 7g of the injection tube main body 7a. These three protrusions 7h, 7i, and 7j are formed as engaged portions, correspondingly to an engaging portion 16b described later, in order to make an engagement between the attachment member 16 and the injection tube 7. The first protrusion 7h is

formed in a state of protruding from one side (left side) surface of the second portion **7g**. The second protrusion **7i** is formed in a state adjacent to and behind the first protrusion **7h**. The second protrusion **7i** protrudes from one side (left side) surface of the second protrusion **7i** with a protrusion amount smaller than that of the first protrusion **7h**. The third protrusion **7j** is formed to protrude downward from the lower surface of the second portion **7g**. The third protrusion **7j** is formed displaced rearward from the first protrusion **7h**.

(66) The nozzle portion **7b** is a portion to be inserted into the incisional wound of the eyeball, when the intraocular lens **4** is injected into the eye using the intraocular lens injector **1**. The nozzle portion **7b** is formed at the tip part of the injection tube **7**. The nozzle portion **7b** is formed so as to protrude forward from the tip of the second portion **7g** of the injection tube main body **7a**. The nozzle portion **7b** is formed in a substantially circular shape when viewed from the axial direction of the injection tube **7**. An outer circumferential diameter of the nozzle portion **7b** is substantially uniform over an entire length **Ln** of the nozzle portion **7b**. The tip part of the nozzle portion **7b** opens with an oblique incisional wound from the top to the bottom, through which the intraocular lens **4** is released to the outside. A cutout portion **7k** is formed in an opening edge on a lower side of the nozzle portion **7b**. The cutout portion **7k** is formed along the axial direction of the injection tube **7**. The cutout portion **7k** is preferably formed in a V shape as shown in the figure.

(67) (Attachment Member)

(68) FIG. **9** is a perspective view showing a configuration of the attachment member. Further, FIGS. **10A** to **10C** a configuration of an attachment member, wherein FIG. **10A** is a plan view, FIG. **10B** is a side view, and FIG. **10C** is a rear view.

(69) As an example of the additional function described above, the attachment member **16** is a member that provides a function of limiting an injection amount of the nozzle portion **7b** (hereinafter also referred to as “nozzle injection amount limiting function”) when the nozzle portion **7b** is inserted into the incisional wound of the eyeball. The nozzle injection amount limiting function is realized (details will be described later) using a tip surface **16d** of the sleeve portion **16a** of the attachment member **16**.

(70) The attachment member **16** is a member separate from the injection tube **7**, and is configured to be movable with respect to the injection tube **7**. A direction and a form of movement of the attachment member **16** with respect to the injection tube **7** are not particularly limited. For example, movement of the injection tube **7** in the axial direction, movement of the injection tube **7** in a direction intersecting (including orthogonal to) the axial direction of the injection tube **7**, movement of the injection tube **7** in a direction around the axis, and movement of the injection tube **7** in a direction around the axis that intersects (including orthogonal to) the axial direction of the injection tube **7**, and the like can be considered as the direction of the movement. Also, linear movement, rotational movement, twist movement, opening and closing (rocking) movement, expansion and contraction movement, and the like can be considered as the form of the movement. In this embodiment, the attachment member **16** is detachably attached to the injection tube **7** by the movement of the attachment member **16** with respect to the injection tube **7**. “Detachably” described here means that “the attachment member **16** can be attached to the injection tube **7** and the attachment member **16** can be detached from injection tube **7**”. In the intraocular lens injector **1** of this embodiment, the attachment member **16** may be provided in a state of being attached to the injection tube **7** or in a state of being detached from the injection tube **7** in some cases. In each case, the attachment member **16** is one of the elements constituting the intraocular lens injector **1**. Further, when using the nozzle injection amount limiting function provided by the attachment member **16**, the attachment member **16** is attached to the injection tube **7**, and when the function is not used, the attachment member **16** is detached from the injection tube **7**. In the latter case, namely, even when the attachment member **16** is not used in the intraocular lens injection surgery, the attachment member **16** is still one of the elements constituting the intraocular lens injector **1**.

(71) The attachment member **16** includes a sleeve portion **16a** and an engaging portion **16b**. A

through hole **16c** is formed in the sleeve portion **16a**, so that the nozzle portion **7b** of the injection tube **7** can be engaged therewith. The through hole **16c** is formed in a state penetrating the sleeve portion **16a** in the axial direction. The tip surface **16d** of the sleeve portion **16a** is formed to be obliquely inclined with respect to the central axis of the sleeve portion **16a**. The tip surface **16d** of the sleeve portion **16a** is disposed in a state of protruding outward (in a direction of enlarging the diameter) from the outer circumferential surface of the nozzle portion **7b** when the attachment member **16** is attached to injection tube **7**. This shows that the tip surface **16d** of the sleeve portion **16a** is a portion corresponding to the “protruding surface”.

(72) The shape of the protruding surface is not limited to an annular flat surface that is continuous in a circumferential direction like the tip surface **16d** of the sleeve portion **16a**, and may be formed in any form as long as it exhibits the nozzle injection amount limiting function. For example, the protruding surface may be discontinuously disposed in the circumferential direction by arrangement at least at one place in the circumferential direction, preferably two places, or three or more places at 180 degrees pitch. Also, the protruding surface may be a curved surface.

(73) The engaging portion **16b** is a portion to be engaged with the injection tube **7** when the attachment member **16** is attached to the injection tube **7**. The engaging portion **16b** is configured so that it can be engaged with the injection tube **7** and can release the engaged state, in order to make the attachment member **16** detachable from injection tube **7**. The engaging portion **16b** has a configuration in which a substantially trapezoidal conical outer circumferential wall corresponding to the outer circumferential shape of the injection tube main body **7a** (second portion **7g**) of the injection tube **7** is partially cut out. A hook portion **16e**, a stopper portion **16f**, and a viewing window **16g** are formed in the engaging portion **16b**. Among them, the hook portion **16e** and the stopper portion **16f** constitute a locking mechanism for temporarily positioning and fixing the attachment member **16** to the injection tube **7**. The hook portion **16e** is a portion to be hooked on the first protrusion **7h** of the injection tube **7**. On the inner circumferential side of the hook portion **16e**, a semicircular recessed portion **16h** is formed. The recessed portion **16h** is configured to be able to engage and disengage with/from the second protrusion **7i** of the injection tube **7** from the direction around the axis of the injection tube **7**. “The direction around the axis” means a direction in which rotation is carried out around the central axis of a certain member. The stopper portion **16f** is configured so that it can abut against the first protrusion **7h** of the injection tube **7** from the direction around the axis of the injection tube **7**, and so that it can abut against the third protrusion **7j** of the injection tube **7** from the axial direction of the injection tube **7**. The viewing window **16g** is formed in a state in which the upper portion of the engaging portion **16b** is cutout in a substantially V shape in plan view. The viewing window **16g** is formed in order to allow a state of the intraocular lens **4** moving inside of the injection tube **7** to be visually recognized from the outside, even when the attachment member **16** is attached to the injection tube **7**. A material of the attachment member **16** is not particularly limited, and for example, metal, ceramic, resin, and the like are used, and a transparent or translucent material is preferable so that the state of the intraocular lens moving inside of the injection tube **7** can be visually recognized from the outside. Further, the protruding surface of the attachment member **16** is in direct contact with a cornea when the intraocular lens is injected into the eye, thereby adding a load on the cornea, and therefore the attachment member **16** is preferably made of a soft resin such as silicone, urethane or the like in order to reduce the load.

(74) FIGS. **11A** to **11C** show a state in which the attachment member is attached to the injection tube, wherein FIG. **11A** is a plan view, FIG. **11B** is a side view, and FIG. **11C** is a bottom view.

(75) As shown in the figure, in a state in which the attachment member **16** is attached to the injection tube **7**, the nozzle portion **7b** is engaged with the through hole **16c** of the sleeve portion **16a**. At this time, the tip surface **16d** of the sleeve portion **16a** is disposed in a state protruding outward from the outer circumferential surface of the nozzle portion **7b**, and is disposed in a state inclined in the same direction as the cutout of the nozzle portion **7b** with respect to the axial

direction of the injection tube 7.

(76) In such a case, an inner diameter of the through hole **16c** of the sleeve portion **16a** may be set to be equal to or slightly larger than the outer circumferential diameter of the nozzle portion **7b** over an entire axial length of the sleeve portion **16a**, in order not to deform the shape of the nozzle portion **7b** engaged with the through hole **16c**. Further, as other preferable embodiment, as shown in FIG. 12A and FIG. 12B, a tapered portion **16i** may be formed in a part of the through hole **16c** of the sleeve portion **16a**. The tapered portion **16i** is formed such that the diameter of the through hole **16c** is gradually decreased toward the tip of the sleeve portion **16a**, on the tip side of the sleeve portion **16a**. A minimum diameter of the through hole **16c** in the tapered portion **16i** is set to be smaller than an outer circumferential diameter of the nozzle portion **7b**. Thereby, in a state in which the attachment member **16** is attached to the insertion tube 7, the tapered portion **16i** comes into contact with the outer circumferential surface of the nozzle portion **7b** and a portion where the cut portion **7k** is formed. Therefore, on the tip side of the sleeve portion **16a**, the outer circumference diameter of the sleeve portion **16a** is reduced by a contact with the tapered portion **16i**.

(77) Meanwhile, the hook portion **16e** of the engaging portion **16b** is hooked on the first protrusion **7h**. At this time, the recessed portion **16h** is in a state of being engaged with the second protrusion **7i** (see FIG. 12A). Further, the stopper portion **16f** of the engaging portion **16b** is in a state of approaching or contacting the third protrusion **7j** of the injection tube 7. At this time, positioning of the attachment member **16** with respect to the injection tube 7 is performed as follows. Namely, in the direction around the axis of the injection tube 7, the position of the attachment member is determined when the recessed portion **16h** is engaged with the second protrusion **7i** of the injection tube 7, and when the hook portion **16e** is hooked on the first protrusion **7h** of the injection tube 7. Further, in the axial direction of the injection tube 7, the position of the attachment member **16** is determined when the stopper portion **16f** abuts on the third protrusion **7j**.

(78) Here, when the attachment member **16** is detached from the insertion tube 7, the engagement state between the attachment member **16** and the insertion tube 7 is released by moving the attachment member **16** in the axial direction of the injection tube 7, after rotating the attachment member **16** in the direction around the axis of the injection tube 7. Specifically, first, the attachment member **16** is turned in a direction of the arrow as shown in FIG. 13A, thereby making the stopper portion **16f** of the engaging portion **16b** brought into contact with or close to the first protrusion **7h** of the injection tube main body **7a** as shown in FIG. 13B. Thereby, the hook portion **16e** of the engaging portion **16b** is disengaged from the first protrusion **7h** of the injection tube 7, and the recessed portion **16h** (see FIGS. 10A to 10C, 12A and 12B) inside of the hook portion **16e** is disengaged from the second protrusion **7i**. Next, as shown in FIG. 13C, the attachment member **16** is pulled out in the direction of the arrow along the axial direction of the injection tube 7. Thereby, the attachment member **16** can be detached from the injection tube 7.

(79) Meanwhile, when the attachment member **16** is attached to the injection tube 7, the attachment member **16** is engaged with the injection tube 7 by a procedure reverse to the above, namely, the attachment member **16** is moved in the axial direction of the injection tube 7, and thereafter the attachment member **16** is rotated and moved in the direction around the axis of the injection tube 7. Specifically, the attachment member **16** is engaged with the injection tube 7 from the direction opposite to the arrow as shown in FIG. 13C, thereby making the stopper portion **16f** of the engaging portion **16b** brought into contact with or close to the third protrusion **7j** of the injection tube 7 as shown in FIG. 13B. Next, by turning the attachment member **16** in the opposite direction to the direction of detachment, the hook portion **16e** of the engaging portion **16b** is hooked on the first protrusion **7h** of the injection tube 7, and the recessed portion **16h** is engaged with the second protrusion **7i**. Thereby, the attachment member **16** can be attached to the injection tube 7. In this way, the attachment member **16** is attached to the injection tube 7, and in this state, the movement of the attachment member **16** with respect to the axial direction of the injection tube 7 is restricted due to the engagement (contact) between the first protrusion **7h** and the hook portion **16e** and due

to the engagement (contact) between the third protrusion **7j** and the stopper portion **16f**. Therefore, the attachment member **16** cannot be moved in the axial direction of the injection tube **7**. In other words, the position of the attachment member **16** is fixed in the axial direction of the injection tube **7**. Further, the engagement state between the engaging portion **16b** of the attachment member **16** and the engaging portion (**7h**, **7i**, **7j**) of the injection tube **7** is not released unless the attachment member **16** attached to the injection tube **7** is rotated in the direction around the axis of the injection tube **7**, because it is impossible to move the attachment member **16** in the axial direction of the injection tube **7** as described above.

(80) (Rotary Member)

(81) As shown in FIG. **14**, the rotary member **8** is attached to the rear end portion of the injector main body **5**. In this attachment state, the rotary member **8** is disposed coaxially with the injector main body **5**, and is rotatably supported in the direction around the axis of the injector main body **5**. The rotary member **8** is formed in a cylindrical shape. The tip end and the rear end of the rotary member **8** are each opened in a circular shape. As shown in FIG. **1** and FIGS. **2A** to **2C**, a plurality of ridges **8a** are formed on the outer circumferential surface of the rotary member **8**. Each of the ridges **8a** is formed to be parallel to the axial direction of the rotary member **8**. The rotary member **8** is a portion rotated by a user when the intraocular lens injector **1** is used by a screw system. At this time, when the plurality of ridges **8a** are formed on the outer circumferential surface of the rotary member **8**, the finger of the user is caught on the ridges **8a**, so that it becomes easy to rotate the rotary member **8**.

(82) First threaded portions **8b** are formed on the inner circumferential surface of the rotary member **8**. The first threaded portions **8b** are formed at a predetermined pitch. Further, two window portions **8c** are formed on the tip part of the rotary member **8**. The two window portions **8c** are formed at intervals of 180 degrees in the circumferential direction. These window portions **8c** are portions for visually checking whether or not the injector main body **5** and the rotary member **8** are properly coupled. A pair of hooking claws **8d** is formed on the inner circumferential portion of the tip of the rotary member **8**. The pair of hooking claws **8d** is hooked on the rotation supporting portion **15** of the injector main body **5** when the rotary member **8** is coupled to the rear end portion of the injector main body **5**. Each of the hooking claws **8d** is formed in the vicinity of the opening on the tip side of the rotary member **8** adjacent to each of the window portions **8c**. The rotary member **8** is rotatably supported in the direction around the axis of the injector main body **5** in a state in which the pair of hook claws **8d** are hooked on the rotation supporting portion **15**. Further, the pair of hook claws **8d** is fitted between the flange portion **14** of the injector main body **5** and the rotation supporting portion **15**. Therefore, the rotary member **8** is freely rotatable around the axis of the injector main body **5** but does not move in the axial direction of the injector main body **5**.

(83) (Plunger)

(84) The plunger **9** is disposed coaxially with the injector main body **5**. The plunger **9** is provided movably in the axial direction of the injector main body **5**. The plunger **9** has a bar shaped plunger shaft **9a**. As shown in FIG. **15**, the tip part of the plunger **9** serves as a seal material attachment portion **9b**. FIG. **15** is a cross-sectional view taken along the line E2-E2 in FIG. **2B**. The seal material attachment portion **9b** is composed of two circular disk portions having an outer circumferential diameter slightly smaller than the inner diameter of the injector main body **5**, and a seal member **17** is attached between the two disk portions. As the seal material **17**, for example, an O ring can be used. When the plunger **9** is moved in the axial direction (frontward) of the injector main body **5**, the seal member **17** is brought into contact with the inner circumferential surface of the injector main body **5**, thereby generating an appropriate sliding resistance.

(85) The longitudinal sectional shape of the plunger shaft **9a** is substantially cruciform as shown in FIG. **16A** and FIG. **16B**. FIG. **16A** is a cross-sectional view taken along the line E3-E3 in FIG. **2B**, and FIG. **16B** is a cross-sectional view taken along the line E4-E4 in FIG. **2B**. As shown in FIG. **1** and FIGS. **2A** to **2C**, second threaded portions **9c** are formed on the upper surface and the lower

surface of the plunger shaft **9a**, respectively. The second threaded portions **9c** are formed corresponding to the first threaded portions **8b** of the rotary member **8**. Therefore, the second threaded portions **9c** are formed at the same pitch as the first threaded portions **8b**. The second threaded portions **9c** are held in a state of constantly meshing with the first threaded portions **8b** of the rotary member **8**. Therefore, when the rotary member **8** is rotated, the plunger **9** moves in the axial direction of the injector main body **5** in accordance with a rotation direction and a rotation amount of the rotary member **8**. The plunger **9** is a portion to be pushed in by the user when using the intraocular lens injector **1** by the push system. Here, the second threaded portions **9c** of the plunger **9** and the corresponding first threaded portions **8b** of the rotary member **8** mesh with each other in a relationship of groove threads and screw threads, and in this meshing portion, the first threaded portions **8b** and the second threaded portions **9c** are inclined at a predetermined taper angle with respect to the Z axis (vertical direction in the figure) as shown in FIG. **14**. When the taper angle of these screw portions **8b**, **9c** is adjusted (set) to an appropriate angle, it is possible to rotate the rotary member **8** with little sense of resistance when the plunger **9** is pushed in.

Particularly, in a case of 5° or more and 15° or less of the taper angle θ at the rear of the second threaded portions **9c** in contact with the first threaded portions **8b** of the rotary member **8** when the plunger **9** is pushed in, the rotary member **8** idles without catching the threaded portions **8b**, **9c**, at the time of a forward movement of the plunger **9** by the pushing operation, which is preferable. Further, in order to efficiently apply a load to the rotary member **8**, which is necessary for moving the plunger **9** forward by the pushing operation, it is desirable to set a groove width (dimension in the Y axis direction) between the threaded portions **9c** of the plunger **9** to 1.0 mm or more and 4.0 mm or less. Also, regarding a screw pitch, it is desirable to adjust the screw pitch in accordance with a release load of an installed intraocular lens.

(86) The tip side of the plunger **9**, including the seal member attachment portion **9b**, is disposed so as to be inserted into the injector main body **5**, and the rear end side of the plunger **9** is disposed so as to protrude rearward from the rotary member **8**. In an initial state before use, a protruding dimension L_p (see FIG. **2B**) of the plunger **9** with a rear end position of the rotary member **8** as a reference, is set to be equal to or larger than a movement dimension of the rod **10** required to push out the intraocular lens **4** from the tip of the injection tube **7**. Here, the initial state before use refers to a state before an operation is performed for pushing out the intraocular lens **4** using the rod **10** when using the intraocular lens injector **1**. In the intraocular lens injector **1** of this embodiment, the rod **10** can be moved forward by either of the rotational operation of the rotary member **8** and the pushing operation of the plunger **9**. Therefore, the initial state before use is the state in which neither the rotational operation of the rotation member **8** nor the pushing operation of the plunger **9** is performed.

(87) In the initial state before use, the plunger **9** is largely drawn out so that the protruding dimension L_p of the plunger shaft **9a** is substantially maximum or close to the maximum. In this state, the seal material attachment portion **9b** of the plunger **9** is disposed in a state of being in proximity to or in contact with the rotation restricting portion **13** of the injector main body **5**. At this time, the seal material attachment portion **9b** of the plunger **9** is opposed to the anti-falloff portions **13d** of the rotation restricting portion **13**. Therefore, when attempting to move the plunger **9** rearward from the initial state before use, the seal material attachment portion **9b** abuts on the anti-falloff portions **13d**. Thereby, fall-off of the plunger **9** from the injector main body **5** is prevented.

(88) Further, when the tip side of the plunger **9** is inserted into the injector main body **5** together with the rod **10**, as shown in FIG. **5** FIGS. **5A** and **5B**, the rotation restricting portion **13** is set in an opened state. Thereby, the tip side of the plunger **9** with the seal member **17** can be inserted into the injector main body **5** without interference with the rotation restricting portion **13**. Further, in this state, when the rotation restricting portion **13** is closed so as to be engaged with the plunger shaft **9a** of the plunger **9**, as shown in FIG. **16A** and FIG. **16B**, a part of the plunger shaft **9a** (a portion

where the second threaded portion 9c is formed) is engaged with the guide groove 13c of the sliding guide 13b. At this time, as shown in FIG. 16B, the two opposing surfaces of the guide groove 13c are disposed so as to support a part of the plunger shaft 9a from both sides. Thereby, the plunger 9 is supported movably in the axial direction of the injector main body 5. However, the rotation of the plunger 9 with respect to the axial direction of the injector main body 5 is restricted by the rotation restricting portion 13.

(89) (Rod)

(90) The rod 10 serves as releasing the intraocular lens 4 from the tip part (opening of the nozzle portion 7b) of the injection tube 7, by pushing out the intraocular lens 4 forward, the intraocular lens 4 being installed on the lens installing portion 11. The rod 10 is formed in an elongated rod shape. The rod 10 is coupled to the tip part of the plunger 9 and moves integrally with the plunger 9 in the axial direction of the hollow body.

(91) <Method of Using the Intraocular Lens Injector>

(92) Next, a method of using the intraocular lens injector 1 will be described. First, the user injects a viscoelastic substance into the injection hole 7c of the injection tube 7. Thereby, the viscoelastic substance is supplied to the intraocular lens 4 installed on the lens installing portion 11 of the injector main body 5.

(93) Next, the user moves the slider 6 forward. Thereby, a lens abutting portion (not shown) formed at the tip part of the slider 6 comes into contact with the intraocular lens 4, and pushes out the intraocular lens 4 as it is. Then, the intraocular lens 4 is deformed into a predetermined shape. When the intraocular lens 4 is deformed by the movement of the slider 6 in this manner, the intraocular lens 4 can be easily folded into a desired shape when the intraocular lens 4 is pushed out by the rod 10 by a subsequent operation.

(94) Next, the user operates the intraocular lens injector 1 by the push system or the screw system to move the rod 10 forward together with the plunger 9, thereby releasing the intraocular lens 4 from the nozzle portion 7b of the injection tube 7. At this time, the rod 10 is brought into contact with the intraocular lens 4 while moving forward, and pushes out the intraocular lens 4 as it is. Then, the intraocular lens 4 moves inside of the injection tube 7 while being folded into a predetermined shape, and is released from the opening at the tip of the nozzle portion 7b. Accordingly, the intraocular lens 4 can be injected into the eye in a small folded state by releasing the intraocular lens 4 in a state in which the nozzle portion 7b of the injection tube 7 is inserted into the incisional wound of the eyeball. Further, after intraocular injection, the intraocular lens 4 can be restored to its original shape by a restoring force of the intraocular lens 4 itself.

(95) <Effect of First Embodiment>

(96) Next, an effect of the intraocular lens injector 1 according to a first embodiment of the present invention will be described.

(97) The intraocular lens injector 1 of this embodiment can cope with two different operation methods, that is, a screw system and a push system. The screw system is a method of pushing out the intraocular lens 4 by the rotational operation of the rotary member 8, and the push system is a method of pushing out the intraocular lens 4 by a pushing operation of the plunger 9. Therefore, in using the intraocular lens injector 1, a user who prefers the screw system can push out the intraocular lens 4 by rotating the rotary member 8, and a user who prefers the push system can push out the intraocular lens 4 by pushing the plunger 9. Each operation method will be described below.

(98) (Screw System)

(99) When the intraocular lens injector 1 is used by the screw system, the user rotates the rotary member 8. Specifically, while rotating the injector body 5 with one hand, the rotary member 8 is rotated with the other hand. At this time, the first threaded portions 8b of the rotary member 8 and the second threaded portions 9c of the plunger 9 are always kept in mesh with each other, including the initial state before use. Therefore, when the rotary member 8 is rotated in a predetermined direction (clockwise direction as seen from the rear end side of the intraocular lens injector 1), the

plunger **9** moves forward in accordance with the rotation of the rotary member **8**, and the rod **10** also moves forward together with the movement of the plunger **9**. Further, since the rotation of the plunger **9** in the direction around the axis of the injector main body **5** is restricted by the rotation restricting portion **13**, the plunger **9** and the rod **10** move forward without rotating in the direction around the axis of the injector main body **5**. Thereby, the intraocular lens **4** installed on the lens installing portion **11** of the injector main body **5**, is pushed out from the tip of the nozzle portion **7b** by the movement of the rod **10**.

(100) (Push System)

(101) When the intraocular lens injector **1** is used by the push system, an operation of pushing the plunger **9** is performed by the user. Specifically, a thumb is pressed against the rear end portion of the plunger **9** while hooking an index finger and a middle finger on the flange portion **14** of the injector main body **5**. Then, in this state, the plunger **9** is pushed forward. Then, a pushing force applied to the plunger **9** is converted to a force for rotating the rotary member **8** by the meshing between the first threaded portions **8b** and the second threaded portions **9c**. Therefore, when the plunger **9** is pushed forward, the plunger **9** moves forward together with the rod **10** and the rotary member **8** rotates in accordance with the movement of the plunger **9**. Further, since the rotation of the plunger **9** in the direction around the axis of the injector main body **5** is restricted by the rotation restricting portion **13**, the plunger **9** and the rod **10** move forward without rotating in the direction around the axis of the injector main body **5**. Thereby, the intraocular lens **4** installed on the lens installing portion **11** of the injector main body **5**, is pushed out from the tip of the nozzle portion **7b** by the movement of the rod **10**.

(102) In this manner, the intraocular lens injector **1** of this embodiment can cope with either one of the operation methods such as the screw system and push system. Accordingly, the user of the intraocular lens injector **1**, can select either one of the operation methods, according to his/her preference, or according to other reasons, circumstances etc. Thereby, with one intraocular lens injector **1**, it is possible to flexibly cope with differences in procedures of an intraocular lens injection surgery. Further, it is possible to cater to both users of the user who prefers the screw system and the user who prefers the push system. Further, the first threaded portions **8b** of the rotary member **8** and the second threaded portions **9c** of the plunger **9**, are always in mesh with each other, including the initial state before use. Therefore, in the case of operating by the screw system, the intraocular lens **4** can be pushed out only by rotating the rotary member **8**. Further, in the case of operating by the push system, the first threaded portions **8b** and the second threaded portions **9c** are held in mesh with each other from the start to the end of the operation. Therefore, vibration or the like is unlikely to occur during the pushing operation of the plunger **9**.

(103) (Nozzle Insertion Amount Limiting Function of the Attachment Member)

(104) According to the intraocular lens injector **1** of this embodiment, it is possible to utilize the nozzle insertion amount limiting function provided by the attachment member **16** or to avoid its use. Specifically, the attachment member **16** is configured to be movable with respect to the injection tube **7**, thereby making it possible to attach and detach the attachment member **16** to/from the injection tube **7**. Therefore, in the state in which the attachment member **16** is attached to the injection tube **7**, the nozzle insertion amount limiting function can be used, and in the state in which the attachment member **16** is detached from the injection tube **7**, use of the nozzle insertion amount limiting function can be avoided. Explanation will be given hereinafter for a case of using the nozzle insertion amount limiting function and a case of not using the nozzle insertion amount limiting function separately.

(105) (Case of Using the Nozzle Insertion Amount Limiting Function)

(106) When using the nozzle insertion amount limiting function provided by the attachment member **16**, the attachment member **16** is set in a state of being attached to the injection tube **7**. Thereby, the nozzle portion **7b** of the injection tube **7** is partially covered with the sleeve part **16a** of the attachment member **16**. The nozzle insertion amount limiting function is exhibited by contact

of the tip surface **16d** of the sleeve part **16a** with the outer surface **50b** of the cornea **50**, when an incisional wound **50a** is formed in the periphery of the cornea **50** of the eyeball, and the nozzle portion **7b** of the injection tube **7** is inserted into the incisional wound **50a** in a cataract surgery as shown in FIG. **17**. At this time, the tip surface **16d** of the sleeve part **16a** functions as a stopper for limiting further insertion of the nozzle portion **7b**, by contact of the tip surface **16d** of the sleeve part **16a** with the outer surface **50b** of the cornea **50**, and by this stopper function, the insertion amount of the nozzle portion **7b** is limited.

(107) Here, when the Wound-assisted method is applied, it is desirable to set a relative position of the injection tube **7** and the attachment member **16**, so that the protruding amount (Maximum value) **L1** of the nozzle portion **7b** with respect to the tip surface **16d** of the sleeve part **16a** is 1.0 mm or more and 1.2 mm or less. Thereby, it is possible to limit the insertion amount of the nozzle portion **7b** with respect to the incisional wound **50a** of the cornea **50** to be equivalent to the protrusion amount **L1**. Accordingly, the intraocular lens **4** can be released from the tip of the nozzle portion **7b** in a state of shallowly inserting the nozzle portion **7b** into the incisional wound **50a** of the cornea **50**. Further, in order to use the nozzle insertion amount limiting function, the movement of the attachment member **16** with respect to the axial direction of the insertion tube **7** is restricted by the engagement between the first protrusion **7h** and the hook portion **16e** and the engagement between the third protrusion **7j** and the stopper portion **16f** in a state in which the attachment member **16** is attached to the injection tube **7**. Therefore, after the intraocular lens **4** is released from the tip of the nozzle portion **7b**, the tip surface **16d** of the attachment member **16** can be separated from the outer surface **50b** of the cornea **50** simultaneously with pulling out the nozzle portion **7b** from the incisional wound **50a** of the cornea **50**. Further, when using the nozzle insertion amount limiting function, as described above, the injection tube **7** and the attachment member **16** are used in a combined state. Therefore, the engaging portion **16b** of the attachment member **16** and the engaging portion (**7h**, **7i**, **7j**) of the injection tube **7** are always maintained in an engagement state, and the engagement state is not canceled.

(108) (Case of not using the nozzle insertion amount limiting function)

(109) Meanwhile, when the nozzle insertion amount limiting function provided by the attachment member **16** is not used, the attachment member **16** is detached from the injection tube **7**. Thereby, the nozzle portion **7b** of the injection tube **7** is in a state in which its entire body is exposed to the outside without being covered by the sleeve portion **16a** of the attachment member **16**. Therefore, when the nozzle portion **7b** of the injection tube **7** is inserted into the incisional wound **50a** of the cornea **50**, the nozzle portion **7b** can be inserted more deeply than in the case of using the nozzle insertion amount limiting function described above. Accordingly, the intraocular lens **4** can be released from the tip of the nozzle portion **7b** in a state in which the nozzle portion **7b** is deeply inserted into the incisional wound **50a** of the cornea **50**.

(110) In this manner, in the intraocular lens injector **1** of this embodiment, the nozzle insertion amount limiting function can be used by attaching the attachment member **16** to the insertion tube **7**, and in addition, use of the nozzle insertion amount limiting function can be avoided by detaching the attachment member **16** from the injection tube **7**. Accordingly, the user of the intraocular lens injector **1**, can use the nozzle insertion amount limiting function or to avoid using it, according to his/her preference, or according to other reasons, circumstances etc. Thereby, even if there are user who prefers the procedure for inserting the nozzle portion **7b** deeply into the incisional wound **50a** of the eye ball and user who prefers the procedure to shallowly insert the nozzle portion **7b**, it is possible to flexibly deal with the preference of each user's technique.

(111) Further, if a plurality of attachment members **16** having different sizes (particularly, lengths) of the sleeve portions **16a** are prepared, it is possible to change (increase or decrease) the protrusion amount of the nozzle portion **7b** with respect to the tip surface **16d** of the sleeve portion **16a**, depending on the size of attached member **16** attached to injection tube **7** when using the nozzle insertion amount limiting function. For example, when it is desired to change the protrusion

amount to a protrusion amount (for example, 5.0 mm or more and 6.5 mm or less) larger than the protrusion amount **L1** of the nozzle portion **7b** shown in FIG. **17**, as shown in FIG. **18**, the protrusion amount (maximum value) **L2** of the nozzle portion **7b** can be secured to be larger than the protrusion amount **L1** by using the attachment member **16** having a size compatible with the above protruding amount, for attachment to the injection tube **7**. Thereby, the insertion amount can be changed when inserting the nozzle portion **7b** into the incisional wound **50a** of the cornea **50**. (112) The intraocular lens injector **1** of this embodiment exhibits the effect of (hereinafter referred to as a “first effect”) enabling selection of the operation method of the intraocular lens injector **1** from the screw system and the push system, and the effect of (hereinafter referred to as a “second effect”) enabling selection as to whether or not to use the nozzle insertion amount limiting function. However, the first effect and the second effect are obtained by different configurations. Namely, the first effect is obtained by a configuration (hereinafter referred to as “first configuration”) including the injector main body **5**, the rotary member **8**, the plunger **9**, and the rod **10** described above, and the second effect is obtained by a configuration (hereinafter referred to as a “second configuration”) including the injection tube **7** and the attachment member **16** described above. Therefore, the first effect can be obtained even without the second configuration and the second effect can be obtained even without the first configuration.

(113) Further, this embodiment employs a configuration in which the outer circumferential diameter of the nozzle portion **7b** can be reduced by forming the tapered portion **16i** in the through hole **16c** of the attachment member **16** and bringing the tapered portion **16i** into contact with the outer circumferential surface of the nozzle portion **7b**. Thereby, even if the size of the incisional wound formed on the eyeball is reduced, the tip part of the nozzle portion **7b** can be easily inserted into the incisional wound. Further, when the intraocular lens **4** is passed through the nozzle portion **7b** in a folded state in a predetermined shape, the nozzle portion **7b** is pressed by the intraocular lens **4** from the inside and is deformed. At this time, if the cutout portion **7k** is formed in the nozzle portion **7b**, even if the nozzle portion **7b** is pressed by the intraocular lens **4** and is deformed, damage such as cracks and the like are less likely to occur in the nozzle portion **7b**. Further, an amount of deformation of the nozzle portion **7b** is suppressed to minimum necessary for releasing the intraocular lens **4**. Therefore, the intraocular lens **4** can be injected from a smaller incisional wound.

(114) Further, this embodiment employs a configuration in which a viewing window **16g** is formed on the attachment member **16**. Therefore, even when the intraocular lens injector **1** is used with the attached member **16** attached to the injection tube **7**, the state of the intraocular lens **4** moving inside of the insertion tube **7** can be visually recognized from the outside through the viewing window **16g**. Accordingly, the user of the intraocular lens injector **1** can operate the intraocular lens injector **1** while visually checking the state of the intraocular lens **4** pushed out by the rod **10**, even when the attachment member **16** is attached.

Second Embodiment

(115) Next, a second embodiment of the present invention will be described.

(116) FIG. **19** is a perspective view showing a state in which the attachment member is attached to the injection tube in the intraocular lens injector according to a second embodiment of the present invention. Hereinafter, the configuration of the injection tube and the attachment member according to the second embodiment of the present invention will be described in detail.

(117) (Injection Tube)

(118) FIG. **20** is a perspective view showing the configuration of the injector tube according to the second embodiment of the present invention.

(119) The injection tube **7** differs from that of the first embodiment only in the configuration of the engaged portion. Namely, the first embodiment employs a configuration in which the engaged portion is configured by three protrusions **7h**, **7i**, **7j**, but the second embodiment employs a configuration in which the engaged portion is configured by one locking claw **7m**. The locking

claw **7m** is formed in a substantially U-shape from both side surfaces to the lower surface of the second portion **7g** of the injection tube main body **7a**, when viewed from the front. Further, the locking claw **7m** is formed to protrude downward from the lower surface of the second portion **7g**.
(120) (Attachment Member)

(121) FIGS. **21A** to **22D** are perspective views showing the configuration of the attachment member according to the second embodiment of the present invention. Further, FIG. **22** shows the configuration of the attachment member according to the second embodiment of the present invention, wherein FIG. **22A** is a plan view, FIG. **22B** is a side view, FIG. **22C** is a bottom view, and FIG. **22D** is a rear view.

(122) The attachment member **16** includes a sleeve portion **16a** and an engaging portion **16b** similarly to the first embodiment, but the configuration of the engaging portion **16b** is different. Namely, in the engaging portion **16b**, a tongue piece **16j** is integrally formed in addition to the viewing window **16g**, but the hook portion **16e** and the stopper portion **16f** are not formed. The tongue piece **16j** is formed so as to extend from the lower portion of the engaging portion **16b** in the axial direction of the attachment member **16**. The tongue piece **16j** is formed in a ladder shape having three engagement holes **16k**. In the following description, the three engagement holes **16k** are distinguished by identification codes such as **16k-1**, **16k-2**, and **16k-3**. The three engagement holes **16k-1**, **16k-2**, and **16k-3** are formed side by side so as to be adjacent to each other with a predetermined interval in the axial direction of the attachment member **16**. Further, the three engagement holes **16k-1**, **16k-2**, and **16k-3** are sequentially disposed from the front toward the rear in the axial direction of the attachment member **16**. The attachment member **16** has a configuration in which the locking claw **7m** of the injection tube **7** is engaged with any one of the three engagement holes **16k-1**, **16k-2**, and **16k-3**, thereby being engaged with the injection tube **7**.

(123) FIGS. **23A** to **23C** show a state in which the attachment member is attached to the injection tube, wherein FIG. **23A** is a plan view, FIG. **23B** is a side view, and FIG. **23C** is a bottom view. Further, FIG. **24A** is a cross-sectional view taken along the line J2-J2 in FIG. **23B**, FIG. **24B** is a cross-sectional view taken along the line J1-J1 in FIG. **23A**, and FIG. **24C** is an enlarged view of portion K in FIG. **24B**.

(124) In the state in which the attachment member **16** is attached to the injection tube **7** as shown in the figure, the nozzle portion **7b** is fitted into the through hole **16c** of the sleeve portion **16a**. Further, the locking claw **7m** formed on the injection tube **7** is engaged with the central engaging hole **16k-2** among the three engagement holes **16k-1**, **16k-2**, and **16k-3** formed on the tongue piece **16j** of the attachment member **16**. At this time, the tip surface **16d** of the sleeve portion **16a** is disposed to protrude outward from the outer circumferential surface of the nozzle portion **7b**, and is disposed in a state inclined in the same direction as the cutout of the nozzle portion **7b** with respect to the axial direction of the injection tube **7**.

(125) When using the nozzle insertion amount limiting function, the locking claw **7m** is engaged with any one of the three engagement holes **16k-1**, **16k-2**, and **16k-3**. In this case, the protrusion amount of the nozzle portion **7b** with respect to the tip surface **16d** of the sleeve portion **16a** is changed depending on which engagement holes **16k** is used for engagement with the locking claw **7m**. FIG. **25A** is a side view showing a state in which the locking claw **7m** is engaged with the rear engagement hole **16k-3**. In this state, the attachment member **16** is disposed relatively closer to the front in the axial direction of the injection tube **7**. Therefore, the protrusion amount of the nozzle portion **7b** becomes relatively small. In contrast, FIG. **25B** is a side view showing a state in which the locking claw **7m** is engaged with the central engagement hole **16k-2**. In this state, the protrusion amount of the nozzle portion **7b** becomes larger in the state of FIG. **25A**. FIG. **25C** is a side view showing a state in which the locking claw **7m** is engaged with the front engaging hole **16k-1**. In this state, the attachment member **16** is disposed relatively closer to the rear side in the axial direction of the injection tube **7**. Therefore, the protrusion amount of the nozzle portion **7b** becomes larger in the state of FIG. **25B**.

(126) As described above, in the second embodiment of the present invention, the protrusion amount of the nozzle portion **7b** can be changed (adjusted) to three levels of large, medium, and small, by appropriately moving the attachment member **16** in the axial direction of the injection tube **7** so that the locking claw **7m** is engaged with one of the engagement holes **16k** of the tongue piece **16j**. Further, if more engagement holes **16k** are formed on the tongue piece **16j** of the attachment member **16**, the protrusion amount of the nozzle portion **7b** can be changed in more stages. Further, if the attachment member **16** is pressed in the axial direction of the injection tube **7** in a state in which the locking claw **7m** of the injection tube **7** is engaged with the central engagement hole **16k-2**, the tongue piece **16j** of the attachment member **16** is elastically deformed (bending deformation) upon receiving a pressing force, and as a result, the locking claw **7m** is engaged with the adjacent engagement hole **16k-1** or **16k-3**. Therefore, the protrusion amount of the nozzle portion **7b** can be changed while the attachment member **16** is attached to the injection tube **7**. Thereby, the protrusion amount of the nozzle portion **7b** can be changed without attaching or detaching the attachment member **16** to or from the injection tube **7**. Therefore, it is possible to save labor for detaching the attachment member **16** from the injection tube **7** and labor for placing the attachment member **16** on a tray or the like. Further, there is no risk of the user inadvertently dropping or losing the attachment member **16** detached from the injection tube **7**.

(127) Further, when the nozzle insertion amount limiting function is not used, for example, the entire body of the nozzle portion **7b** can be exposed by releasing the engagement state between the locking claw **7m** and the tongue piece **16j** and pulling out the attachment member **16** from the injection tube **7**. Alternatively, it is also acceptable that the entire body of the nozzle portion **7b** is exposed when the attachment member **16** is moved backward by a predetermined amount while the attachment member **16** is attached to the injection tube **7**, and in this state, the attachment member **16** is engaged with the injection tube **7** by the engagement between the locking claw **7m** and the engagement hole **16k**. When this configuration is employed, it is possible to switch whether or not to use the nozzle insertion amount limiting function without detaching the attachment member **16** from the injection tube **7**. Therefore, the attachment member **16** is not required to be detachable from the injection tube **7**.

Third Embodiment

(128) FIG. **26** is a perspective view showing a configuration of the intraocular lens injector according to a third embodiment of the present invention. Further, FIG. **27A** is a plan view showing the configuration of the intraocular lens injector according to the third embodiment of the present invention, wherein FIG. **27B** is a side view, and FIG. **27C** is a cross-sectional view taken along the line M-M in FIG. **27A**.

(129) In the intraocular lens injector **1** according to the third embodiment of the present invention, a position and a configuration of the rotation restricting portion **13**, a configuration of part of the plunger **9**, a configuration of the injection tube **7**, and a configuration of the attachment member **16** are different.

(130) (Position and Configuration of the Rotation Restricting Portion)

(131) FIGS. **28A** and **28B** are perspective views showing a configuration of the injector main body according to the third embodiment of the present invention, FIG. **28A** shows a state in which the rotation restricting portion provided in the injector main body is opened, and FIG. **28B** shows a state in which the rotation restricting portion is closed.

(132) The rotation restricting portion **13** is formed at the rear end portion of the injector main body **5** located slightly rearward of the flange portion **14** in the axial direction of the injector main body **5**. When the rotary member **8** is attached to the rear end portion of the injector main body **5**, the rotation restricting portion **13** is disposed inside of the rotary member **8** (see FIG. **27C**). Therefore, the entire body of the rotation restricting portion **13** is shielded by the rotary member **8**. The function, the configuration, and the like of the rotation restricting portion **13** are basically the same as those of the first embodiment.

(133) As described above, by employing a configuration in which the rotation restricting portion **13** is formed at the rear end portion of the injector main body **5** and the rotation restricting portion **13** is shielded by the rotary member **8** attached thereto, it is possible to shorten a length of the injector main body **5** and reduce a size of the intraocular lens injector **1**, compared to the first embodiment. Further, after assembling the intraocular lens injector **1**, the rotation restricting portion **13** is hidden inside of the rotary member **8**, and therefore the rotation restricting portion **13** cannot be opened. Therefore, it is possible to reduce the risk of decomposing the intraocular lens injector **1** by a surgeon or the like. Further, it is not necessary to form the rotation restricting portion **13** in front of the flange portion **14**, and therefore it is possible to reinforce the flange portion **14** by forming reinforcing ribs **18** on the outer circumferential surface of the injector main body **5**. Thereby, even when a strong force is applied to the flange portion **14** by the operation by the push system, deformation of the flange portion **14** can be suppressed.

(134) Note that the third embodiment employs a configuration in which the entire body of the rotation restricting portion **13** is shielded by the rotary member **8**. However, the present invention is not limited thereto, and it is possible to reduce the risk of decomposing the intraocular lens injector **1** by a surgeon or the like, as long as at least a part of the rotation restricting portion **13** is shielded by the rotary member **8**. Further, the position of the rotation restricting portion **13** employed in the third embodiment can also be employed in the first embodiment and the second embodiment.

(135) Further, the third embodiment employs a hook mechanism, in order to hold the rotation restricting portion **13** in a closed state. For example as shown in FIG. **39A**, the hook mechanism includes a first hook claw **13e** formed in the rotation restricting portion **13** and a second hook claw **19** formed on the outer circumferential surface of the injector main body **5**. FIGS. **39A** to **39D** each shows a case in which the rotation restricting portion **13** is viewed from behind the injector main body **5**.

(136) The first hook claw **13e** is formed on the opposite side of the pivotally supporting portion **13a** in the rotation restricting portion **13**. The pivotally supporting portion **13a** connects the rotation restricting portion **13** and the injector main body **5** and supports the rotation restricting portion **13** so as to be rotatable (openable and closable). The second hook claw **19** is formed on the opposite side of the pivotally supporting portion **13a** in the left-right direction.

(137) In the hook mechanism having the above configuration, when the rotation restricting portion **13** is closed from the opened state, the first hook claw **13e** of the rotation restricting portion **13** is caught by the second hook claw **19** of the injector main body **5** and locked. At this time, in a radial direction of the injector main body **5**, the first hook claw **13e** is disposed relatively outside, and the second hook claw **19** is disposed relatively inside. Further, the plunger **9** is fitted into a guide groove **13c** formed between the pair of sliding guides **13b** of the rotation restricting portion **13**. Therefore, when the rotary member **8** is rotated so as to use the intraocular lens injector **1** by the screw system, the rotation of the plunger **9** in the direction around the axis of the injector main body **5** is restricted by the rotation restricting portion **13**.

(138) Further, since the first threaded portions **8b** of the rotary member **8** and the second threaded portions **9c** of the plunger **9** are meshed with each other. Therefore, when the rotary member **8** is rotated so as to move the plunger **9** forward, a rotational force F_r is applied to the plunger **9** by the rotation of the rotary member **8**. When the rotational force F_r is applied to the plunger **9**, the rotation restricting portion **13** is pushed by the plunger **9**. As a result, the first hook claw **13e** of the rotation restricting portion **13** is pressed against the second hook claw **19**. Therefore, it is possible to hold the first hook claw **13e** and the second hook claw **19** in a locked state. Accordingly, there is no possibility that the first hook claw **13e** and the second hook claw **19** are disengaged during the rotational operation of the rotary member **8**.

(139) In contrast, for example as shown in FIG. **39B**, when a positional relationship between the pivotally supporting portion **13a** of the rotation restricting portion **13** and the first hook claw **13e** is right/left reversed, and when the second hook claw **19** is formed on the injector main body **5** in

accordance with the reversed positional relationship, there is a possibility that the first hook claw **13e** and the second hook claw **19** are disengaged during the rotational operation of the rotary member **8**. Specifically, when the rotational force F_r is applied to the plunger **9** by the rotational operation of the rotary member **8**, there is a case that the rotation restricting portion **13** is deformed or displaced by being pushed by the rotational force F_r , depending on the characteristics of a material constituting the injector main body **5** (including the rotation restricting portion **13**). As a result, there is a possibility that the first hook claw **13e** is displaced outwardly and disengaged from the second hook claw **19**. Therefore, as shown in FIG. **39A**, it is preferable that the rotational force F_r applied to the plunger **9** by the rotational operation of the rotary member **8** is applied in a direction to press the first hook claw **13e** against the second hook claw **19**.

(140) Further, as shown in FIG. **39C**, it is also possible to employ a configuration in which the first hook claw **13e** of the rotation restricting portion **13** is disposed relatively inside and the second hook claw **19** of the injector main body **5** is disposed relatively outside. In this configuration as well, the rotational force F_r applied to the plunger **9** by the rotational operation of the rotary member **8** is applied in a direction to press the first hook claw **13e** against the second hook claw **19**. Therefore, there is no possibility that the first hook claw **13e** and the second hook claw **19** are disengaged during the rotational operation of the rotary member **8**. Incidentally, as shown in FIG. **39D**, when the positional relationship between the pivotally supporting portion **13a** of the rotation restricting portion **13** and the first hook claw **13e** is right/left reversed, and when the second hook claw **19** is formed on the injector main body **5** in accordance with the reversed positional relationship, there is a case that the rotation restricting portion **13** is deformed or displaced by being pushed by the rotational force F_r applied to the plunger **9**. As a result, there is a possibility that the first hook claw **13e** is pushed outwardly and disengaged from the second hook claw **19**. Therefore, it is preferable to employ the configuration shown in FIG. **39C**.

(141) Note that the hook mechanism shown in FIGS. **39A** to **39D** can be applied to the first embodiment and the second embodiment described above, and in this case as well, it is preferable to employ the configuration shown in FIG. **39A** or FIG. **39C**.

(142) (Configuration of the Plunger)

(143) A pressing plate portion **9d** is formed at the rear end portion of the plunger **9**. The pressing plate portion **9d** is a portion for pressing a thumb of a surgeon in a case of the operation by the push system, and is formed so as to protrude radially more than the plunger shaft **9a**. In the pressing plate portion **9d**, unevenness may be provided on the surface against which the surgeon's thumb is pressed, for anti-falloff purpose.

(144) When the pressing plate portion **9d** is provided at the rear end portion of the plunger **9** as described above, the operation by the push system becomes easy. Further, the rear end portion of the plunger **9** comes into contact with the thumb on a wider surface compared with the first embodiment. Therefore the load applied to the thumb at the time of pushing the plunger **9** is dispersed. Therefore, a burden on the thumb can be reduced.

(145) The configuration in which the pressing plate portion **9d** is provided at the rear end portion of the plunger **9**, can be employed in the first embodiment and the second embodiment described above.

(146) (Configuration of the Injection Tube)

(147) FIGS. **29A** to **29C** show a configuration of the injection tube according to the third embodiment of the present invention, wherein FIG. **29A** is a plan view, FIG. **29B** is a side view, and FIG. **29C** is a bottom view. Further, FIG. **30A** is a perspective view of the injection tube according to the third embodiment of the present invention as seen obliquely from above, and FIG. **30B** is a perspective view as seen diagonally from below.

(148) The injection tube **7** has a hollow injection tube main body **7a** and a narrow tubular nozzle portion **7b**, and an injection hole **7c** is formed on the upper surface of the injection tube main body **7a**. The injection tube main body **7a** is divided into a first portion **7f** and a second portion **7g**.

Further, the rear end portion of the injection tube main body **7a** is opened, and a hook portion **7d** and a wedge portion **7e** are formed around the opening portion. A cutout portion **7k** is formed in the nozzle portion **7b**. The above points are similar to those of the first embodiment.

(149) An engaged portion **71** is formed on a lower surface side of the first portion **7f** of the injection tube main body **7a**. Two small protrusions **71a**, **71b** and a recessed groove **71c** are formed in the engaged portion **71**. The small protrusion **71b** is disposed between the small protrusion **71a** and the recessed groove **71c** in the axial direction of the injection tube **7**. The small protrusion **71a** has a slope **71d**. The slope **71d** has an inclination such that the protrusion amount of the small protrusion **71a** is gradually increased toward the tip side of the injection tube **7**.

(150) A pair of right and left guide ribs **72** is formed, on both sides of the second portion **7g** of the injection tube main body **7a**. The pair of guide ribs **72** performs a guide function for relatively positioning the injection tube **7** and the attachment member **16** in the direction around the axis of the injection tube **7** when the attachment member **16** is attached to the injection tube **7**. The pair of guide ribs **72** is formed so as to protrude like wings from both sides of the injection tube main body **7a** (the second portion **7g**).

(151) (Configuration of the Attachment Member)

(152) FIGS. **31A** and **31B** are perspective views of the attachment member according to a third embodiment of the present invention as seen obliquely from above, FIG. **31A** shows the attachment member before being attached to the injection tube, and FIG. **31B** shows the attachment member after being attached to the injection tube. Further, FIGS. **32A** to **32C** show a configuration of the attachment member according to the third embodiment of the present invention, wherein FIG. **32A** is a plan view, FIG. **32B** is a side view, and FIG. **32C** is a bottom view. Further, FIG. **33** is a view as seen from arrow N in FIG. **32B**. Note that FIGS. **32A** to **32C** show the attachment member before being attached to the injection tube.

(153) As in the first embodiment, the attachment member **16** is a member that provides the nozzle insertion amount limiting function. The material of the attachment member **16** is preferably a resin, more preferably a transparent or translucent resin. The attachment member **16** is desirably made of an integrally molded product of resin. The attachment member **16** is a member separate from the injection tube **7**, and is configured to be movable with respect to the injection tube **7**. A movable direction of the attachment member **16** with respect to the injection tube **7** is an axial direction of the injection tube **7**. Further, the third embodiment is configured as follows: although the attachment member **16** cannot be detached from the injection tube **7** after the attachment member **16** is attached to the injection tube **7**, the attachment member **16** can move in the axial direction of the injection tube **7** while the attachment member **16** is attached to the injection tube **7** (details will be described later).

(154) A tip surface **161** as a protruding surface is formed at the tip part of the attachment member **16**. The protruding surface of the attachment member **16** is a portion that directly comes into contact with a cornea when the intraocular lens is injected into the eye. Therefore, in order not to damage the cornea as much as possible, a body portion (excluding the portion of the protruding surface) of the attachment member **16** may be formed of hard resin, and the portion of the protruding surface may be formed of a soft resin such as silicone or urethane. The tip surface **161** as the protruding surface is formed in a shape (C shape) in which the upper part of an annular ring is cut out.

(155) A viewing window **162** is formed in the upper part of the attachment member **16**. A punched hole **163** is formed on the bottom of the attachment member **16**. The viewing window **162** is formed to allow the state of the intraocular lens **4** moving inside of the injection tube **7** to be visually recognized from the outside, even when the attachment member **16** is attached to the injection tube **7**. A width of the viewing window **162** is gradually narrowed from the rear end side toward the tip side of the attachment member **16** in accordance with a change in the width of the injection tube **7**. The tip side of the viewing window **162** is opened without being closed and

communicates with a cutout portion of the tip surface **161**. Therefore, even when the attachment member **16** is attached to the injection tube **7**, the whole part from the injection tube main body **7a** of the injection tube **7** to the nozzle portion **7b** can be viewed from the outside through the viewing window **162**. Accordingly, when the intraocular lens **4** is pushed out through the nozzle portion **7b** of the injection tube **7**, it is possible to visually recognize the state of the intraocular lens **4** moving inside of the injection tube **7** without being disturbed by the attachment member **16**.

(156) A pair of left and right side plate portions **164** are provided on both sides of the attachment member **16**. As shown in FIG. **34**, a slit groove **164a** is formed on the inner surface of each side plate portion **164**. The slit groove **164a** is formed corresponding to the guide rib **72** of the injection tube **7**. Note that FIG. **34** is a longitudinal sectional view of the attachment member according to a third embodiment of the present invention, showing a state in which the engaging portion **167** is closed. Grip portions **165** are formed on the outer surface of each side plate portion **164**. The grip portions **165** are provided on the left and right sides of the attachment member **16** in order to be able to grip the attachment member **16** with an index finger and a thumb when the surgeon or the like handles the attachment member **16**. Each grip portion **165** has a concavo-convex structure so that a surgeon or the like can grip the attached member **16** easily.

(157) A bridge portion **166** and an engaging portion **167** are provided at the rear portion of the attachment member **16**. The bridge portion **166** is formed at an upper portion of the attachment member **16** so as to bridge between the pair of side plate portions **164**. A part of the bridge portion **166** is formed by being bent into a waveform so as not to block the injection hole **7c** when the attachment member **16** is attached to the injection tube **7**. The bridge portion **166** is connected to the engaging portion **167** by a connecting portion **168**. The connecting portion **168** is provided in one of the left and right directions. The connecting portion **168** is formed thin so as to have moderate flexibility. The engaging portion **167** is rotatably supported around the connecting portion **168** at the center (fulcrum) by utilizing the flexibility of the connecting portion **168**. FIG. **33** shows a state in which the engaging portion **167** is opened, and from this state, by rotating the engaging portion **167** in a direction indicated by the two-dot chain line arrow in the figure, the engaging portion **167** can be closed. When the attachment member **16** is attached to the injection tube **7**, by rotating the engaging portion **167** so as to close it, the engaging portion **167** can be engaged with the engaged portion **71**.

(158) A coupling claw **169** is formed in the bridge portion **166**. The coupling claw **169** is formed to protrude on the side opposite to the connecting portion **168** in the left-right direction. Meanwhile, a coupling hole **170** corresponding to the coupling claw **169** is formed in the engaging portion **167**. The coupling claw **169** and the coupling hole **170** serve as annularly coupling the bridge portion **166** and the engaging portion **167**, by connecting the bridge portion **166** and the engaging portion **167** on the side opposite to the connecting portion **168**. Specifically, the coupling claw **169** and the coupling hole **170** are engaged with each other and connected by rotating the engaging portion **167** so as to be closed, with the connecting portion **168** as a center. Thereby, the bridge portion **166** and the engaging portion **167** can be annularly coupled. When the attachment member **16** is attached to the injection tube **7**, the bridge portion **166** is disposed on the upper side of the injection tube main body **7a** (the first portion **7f**), and the engaging portion **167** is disposed on the lower side of the injection tube main body **7a**. Therefore, the injection tube main body **7a** is surrounded by the bridge portion **166** and the engaging portion **167**. Further, in a state in which the bridge portion **166** and the engaging portion **167** are coupled, their inner circumferential surfaces have a shape corresponding to the outer circumferential surface of the injection tube main body **7a**.

(159) The engaging portion **167** has a fixing piece **171** and a movable piece **172** extending forward from the fixing piece **171**. A hole **173** is formed between the movable piece **172** and the fixing piece **171**. The movable piece **172** has a bending property, with the connecting portion between the fixed piece **171** and the movable piece **172** as a fixed end and the tip part **172a** of the movable piece **172** as a free end (referred to as “leaf spring property” hereafter). The fixing piece **171** and

the movable piece **172** are formed corresponding to the small protrusions **71a** and **71b** and the recessed groove **71c** constituting the engaged portion **71** of the injection tube **7**. The tip part **172a** of the movable piece **172** is configured to be able to get over the small protrusion **71a** by moving along the slope **71d** of the small protrusion **71a**. Then, the tip part **172a** of the movable piece **172** gets over the small protrusion **71a** along the slope **71d**, whereby the attachment member **16** moves forward, and the protrusion amount of the nozzle portion **7b** is changed.

(160) A coupling bar **174** is formed at a lower portion of the attachment member **16** so as to bridge between the pair of side plate portions **164**. The coupling bar **174** is disposed so as to be opposed to the tip part **172a** of the movable piece **172** when the engaging portion **167** is closed. In the axial direction of the attachment member **16**, a punched hole **163** is formed in front of the coupling bar **174**.

(161) (Procedure for Attaching the Attachment Member)

(162) Next, a procedure for attaching the attachment member **16** to the injection tube **7** will be described. Note that attachment of the attachment member **16** is performed in a manufacturing process (assembly process) of the intraocular lens injector **1**.

(163) First, as shown in FIG. **35**, the attachment member **16** with the engaging portion **167** opened, is disposed in front of the nozzle portion **7b** of the injection tube **7**.

(164) Next, as shown in FIG. **36**, by bringing the attachment member **16** and the injection tube **7** relatively closer to each other, the attachment member **16** is put on the outside of the injection tube **7**. At this time, the attachment member **16** is inserted to the rear end portion of the injection tube **7**. Then, the nozzle portion **7b** protrudes from the tip surface **161** of the attachment member **16** by a predetermined amount. Further, when the attachment member **16** is inserted into the injection tube **7**, a pair of guide ribs **72** formed on the injection tube **7** is engaged with a pair of slit grooves **164a** formed in the attachment member **16** so as to correspond to the pair of guide ribs **72**. Thereby, the injection tube **7** and the attachment member **16** are positioned in the direction around the axis of the injection tube **7**.

(165) Next, by rotating the engaging portion **167** so as to be closed, with the connecting portion **168** of the attached member **16** as a center, the coupling hole **170** of the engaging portion **167** and the coupling claw **169** of the bridge portion **166** are fitted and connected to each other. Thereby, the bridge portion **166** and the engaging portion **167** are annularly coupled.

(166) With the above procedure, the attachment member **16** can be attached to the injection tube **7**. The attachment member **16** cannot be detached from the injection tube **7** after the attachment member **16** is attached to the injection tube **7**. Namely, in the third embodiment of the present invention, the attachment member **16** cannot be detached from the injection tube **7**. However, the attachment member **16** is configured to be movable with respect to the injection tube **7**. This point will be described hereafter.

(167) First, as described above, when the attachment member **16** is attached to the injection tube **7**, the positional relationship between the small protrusion **71a**, the small protrusion **71b**, the recessed groove **71c** of the injection tube **7**, and the fixing piece **171**, the movable piece **172**, the coupling bar **174** of the attachment member **16**, is as shown in FIG. **37**. Namely, the small protrusion **71a** is disposed between the tip part **172a** of the movable piece **172** and the coupling bar **174**, and the small protrusion **71b** is disposed between the tip part **172a** of the movable piece **172** and the fixing piece **171**, in the axial direction of the injection tube **7**. Further, the tip part **172a** of the movable piece **172** is disposed in contact (or close proximity) to a front surface of the small protrusion **71b** adjacent to the slope **71d** of the small protrusion **71a**, and the coupling bar **174** is disposed in contact (or close proximity) to an entire surface of the small protrusion **71a**.

(168) When the attachment member **16** is moved from the above state, first, a surgeon or the like grips a pair of gripping portions **165** formed on both sides of the attachment member **16** with two fingers (usually an index finger and a thumb). Next, a force in a direction **P** is applied to the attachment member **16** while gripping the pair of gripping portions **165** with fingers. Then, the

movable piece **172** formed at the lower part of the attachment member **16**, is deformed due to the leaf spring property of the movable piece **172** itself, while contacting the slope **71d** of the small protrusion **71a** of the injection tube **7**. Then, the tip part **172a** of the movable piece **172** moves along the slope **71d**, and the movable piece **172** returns to an original shape due to its own leaf spring property when getting over the protruding end portion of the small protrusion **71a**.

(169) Thereby, the positional relationship between the small protrusion **71a**, the small protrusion **71b**, the recessed groove **71c** of the injection tube **7**, and the fixing piece **171**, the movable piece **172**, the coupling bar **174** of the attachment member **16**, is as shown in FIG. **38**. Namely, the small protrusion **71a** is disposed between the tip part **172a** of the movable piece **172** and the fixing piece **171**, and the small protrusion **71b** is also disposed between the tip part **172a** of the movable piece **172** and the fixing piece **171** in the axial direction of the injection tube **7**. Further, the tip part **172a** of the movable piece **172** is disposed in contact (or close proximity) to the front surface of the small protrusion **71a**, and the fixing piece **171** is disposed in contact (or close proximity) to the rear surface of the small protrusion **71b**.

(170) As described above, in this embodiment, the attachment member **16** can move in the axial direction of the injection tube **7**, while the attachment member **16** is attached to the injection tube **7**. Therefore, the protrusion amount of the nozzle portion **7b** can be changed, while the attachment member **16** is attached to the injection tube **7**. Thereby, the surgeon who prefers the procedure for inserting the nozzle portion **7b** deeply into the incisional wound of an eyeball, uses the intraocular lens injector **1** in a state shown in FIG. **37**, and the surgeon who prefers the procedure for shallowly inserting the nozzle portion **7b**, can use the intraocular lens injector **1** in a state shown in FIG. **38**. Accordingly, the surgeon performing the cataract surgery can selectively use the intraocular lens injector **1** according to the procedure of his/her choice.

(171) Note that the third embodiment employs the configuration in which the protrusion amount of the nozzle portion **7b** can be adjusted in two stages by allowing the attachment member **16** to move in the axial direction of the injection tube **7**. However, the present invention is not limited to this configuration, and it is also acceptable to employ a configuration in which the protrusion amount of the nozzle portion **7b** can be adjusted in multiple stages of three or more stages.

(172) Further, the third embodiment employs the configuration in which the attachment member **16** cannot be returned to the position (backward) before the movement, due to the contact between the tip part **172a** of the movable piece **172** and the small protrusion **71a**, after the attachment member **16** attached to the injection tube **7** is moved forward. However, the present invention is not limited to this configuration, and it is also acceptable to employ a configuration in which the attachment member **16** can be returned to its original position.

Modified Example, Etc

(173) The technical scope of the present invention is not limited to the embodiment described above but includes various modes and modifications as far as the specific effects obtained by the constituent features of the invention and combinations thereof can be derived.

(174) For example, in the above embodiment, the plunger **9** and the rod **10** are formed as separate members, but they may be formed as an integral structure.

(175) Further, in the above embodiment, the viewing window **16g** is formed in a state in which a part of the attachment member **16** is cut out. However, the present invention is not limited thereto, and a hole (not shown) having a proper size may be formed in a part of the attachment member **16**, and this hole may be used as a viewing window.

(176) Further, in the above embodiment, the injector main body **5** and the injection tube **7** are mutually assembled to form a hollow body. However, the hollow body may have an integral structure (such as an integrally molded product of resin).

(177) Further, in the above embodiment, the slider **6** is included in one of the components of the intraocular lens injector **1**. However, the slider **6** is an auxiliary member for folding the intraocular lens **4** into a desired shape, and is not always necessary to inject the intraocular lens **4** into the eye.

Therefore, it is not necessary to provide the slider **6**, as long as the intraocular lens **4** can be folded into a desired shape merely by pushing out the intraocular lens **4** using the rod **10**, for example by devising the configuration of the lens installing portion **11** in the injector main body **5** and the configuration of the tip of the rod **10**.

DESCRIPTION OF SIGNS AND NUMERALS

(178) **1** Intraocular lens injector **4** Intraocular lens **5** Injector main body **7** Injection tube **7b** Nozzle portion **7k** Cutout portion **7m** Locking claw **8** Rotary member **9** Plunger **9c** Second threaded portion **10** Rod **11** Lens installing portion **13** Rotation restricting portion **13d** Anti-falloff portion **16** Attachment member **16a** Tip surface **16b** Engaging portion **16g** Viewing window **16i** Tapered portion **16j** Tongue piece **16k** Engagement hole

Claims

1. An intraocular lens injector for injecting an intraocular lens into an eye, comprising: a hollow body having a nozzle portion configured to be inserted into an incisional wound in the eye and defining an outer circumferential surface and a distal end with a distal end opening; an attachment member including a sleeve portion defining an axis and a distal end and having a protruding surface at the distal end that is oriented at an oblique angle with respect to the axis and is configured to engage the outer surface of the eye when the nozzle portion is within the incisional wound; and a protrusion and hole arrangement associated with the hollow body and the attachment member that is configured such that the attachment member cannot be detached from the hollow body after the attachment member has been attached to the hollow body; wherein the protrusion and hole arrangement is configured such that the attachment member has an engaged state where the attachment member is fixed to the hollow body at a first location and a disengaged state where the attachment member is movable with respect to the hollow body to a second location axially offset from the first location; and wherein the protruding surface at the distal end of the attachment member is located rearward of the distal end of the nozzle portion and protrudes outwardly from the nozzle portion when the attachment member is in the engaged state.
2. The intraocular lens according to claim 1, wherein the attachment member is movable while being attached to the hollow body.
3. The intraocular lens injector according to claim 1, wherein the protrusion and hole arrangement comprises an engaging portion which is engaged with the hollow body and is formed in the attachment member, and an engaged portion which is formed in the hollow body corresponding to the engaging portion.
4. The intraocular lens injector according to claim 3, wherein the engaging portion has a lock mechanism in which the attachment member is temporarily fixed to the hollow body by an engagement between the engaging portion and the engaged portion.
5. The intraocular lens injector according to claim 3, wherein the engaging portion includes a tongue piece having a plurality of engagement holes formed thereon, and the engaged portion includes a locking claw engaged with one of the plurality of engagement holes.
6. The intraocular lens injector according to claim 1, wherein the protruding surface restricts an injection amount when the nozzle portion is within the incisional wound.
7. The intraocular lens injector according to claim 6, wherein the attachment member protruding surface is soft.
8. The intraocular lens injector according to claim 6, wherein the attachment member protruding surface is formed from a soft resin.
9. The intraocular lens injector according to claim 1, wherein a viewing window is provided on the attachment member.
10. The intraocular lens according to claim 9, wherein the protruding surface is formed in a shape with an upper part cut out, and the viewing window is formed so as to communicate with a cutout

portion of the protruding surface.

11. The intraocular lens injector according to claim 1, wherein the protruding surface is made of a soft resin.

12. The intraocular lens injector according to claim 1, wherein the nozzle portion has a cutout portion formed along an axial direction of the hollow body and inclined in the same direction as the protruding surface of the attachment member, and a tapered portion is formed on the attachment member, to reduce an outer circumferential diameter of the nozzle portion, by being brought into contact with an outer circumferential surface of the nozzle portion, which is a portion where the cutout portion is formed.

13. The intraocular lens injector according to claim 1, wherein the hollow body has a lens installing portion, and an intraocular lens is installed on the lens installing portion.

14. The intraocular lens injector according to claim 1, wherein the engaged state where the attachment member is fixed to the hollow body at the first location defines a first engaged state; and the protrusion and hole arrangement is configured such that the attachment member is in a second engaged state at the second location.

15. An intraocular lens injector for injecting an intraocular lens into an eye, comprising: a hollow body having a main body and an injection tube, secured to the main body, with a nozzle portion configured to be inserted into an incisional wound in the eye and defining an outer circumferential surface and a distal end; an attachment member defining a distal end and having a soft protruding surface at the distal end configured to engage the outer surface of the eye when the nozzle portion is within the incisional wound; a protrusion and hole arrangement associated with attachment member and the injection tube that is configured such that the attachment member has an engaged state where the attachment member is fixed to the injection tube at a first location and a disengaged state where the attachment member is movable with respect to the injection tube to a second location axially offset from the first location.

16. The intraocular lens injector according to claim 15, wherein wherein the attachment member includes a sleeve portion defining an axis and the protruding surface is oriented at an oblique angle with respect to the axis.

17. The intraocular lens injector according to claim 15, wherein the attachment member protruding surface is formed from a soft resin.

18. The intraocular lens injector according to claim 15, wherein the attachment member cannot be detached from the hollow body.

19. The intraocular lens injector according to claim 15, wherein the engaged state where the attachment member is fixed to the injection tube at the first location defines a first engaged state; and the protrusion and hole arrangement is configured such that the attachment member is in a second engaged state at the second location.

20. The intraocular lens injector according to claim 15, wherein the protrusion and hole arrangement comprises a plurality of protrusions and a plurality of holes.

21. The intraocular lens injector according to claim 15, wherein the distal end of the nozzle portion defines the distal end of the intraocular lens injector; and the attachment member is located proximal of the distal end of the nozzle portion and protrudes outwardly from the nozzle portion when the attachment member is in the engaged state.

22. An intraocular lens injector for injecting an intraocular lens into an eye, comprising: a hollow body having a main body and an insertion tube with a nozzle portion that is configured to be inserted into an incisional wound in the eye and that defines an outer circumferential surface and a distal end with a distal end opening; a rod configured to push an intraocular lens through the nozzle portion; a plunger configured to move the rod forward; and an attachment member defining a distal end and having a protruding surface at the distal end; wherein the respective configurations of the hollow body, the rod and the plunger are such that the hollow body, the rod and the plunger are capable of together functioning to drive the intraocular lens through the nozzle portion and into an

eye in the absence of the attachment member; wherein the attachment member and the hollow body are respectively configured such that the attachment member is movable from a first stage where the attachment member is fixed to the hollow body to at least a second stage, which is axially offset from the first stage, where the attachment member is fixed to the hollow body; and wherein the protruding surface at the distal end of the attachment member is located rearward of the distal end of the nozzle portion and protrudes outwardly from the nozzle portion when the attachment member is in the first and second stages.

23. The intraocular lens injector according to claim 22, wherein the attachment member and the hollow body are respectively configured such that the attachment member is movable from the second stage to a third stage, which is axially offset from the second stage, where the attachment member is fixed to the hollow body.

24. The intraocular lens injector according to claim 22, wherein the attachment member includes first and second engagement holes and the hollow body includes a protrusion that is received in the first engagement hole when the attachment member is in the first stage and is received in the second engagement hole when the attachment member is in the second stage.

25. The intraocular lens injector according to claim 22, wherein the hollow body includes first and second protrusions; the attachment member includes a fixing piece, a movable piece that includes a tip part, and a hole between the movable piece and the fixing piece; the tip part of the movable piece is between the first and second protrusions when the attachment member is in the first stage; and the first and second protrusions are between the tip part of the movable piece and the fixing piece when the attachment member is in the second stage.
